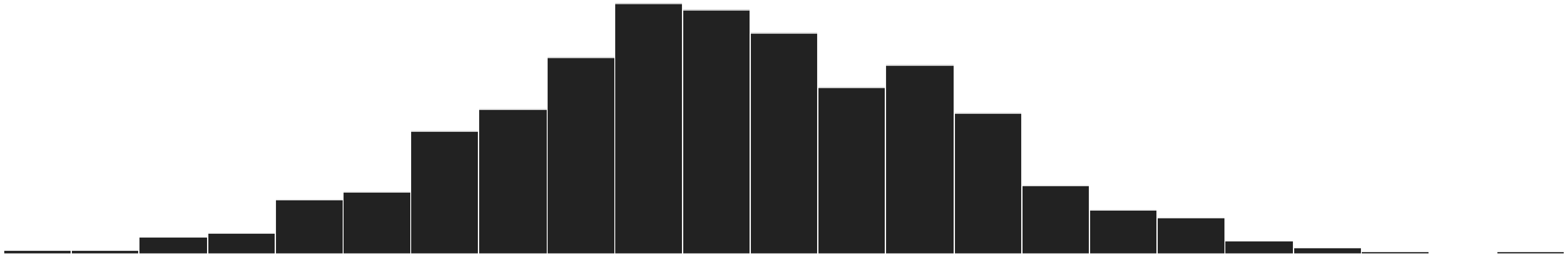
VTPEH 6105

Lecture 04

Hypothesis Testing: Comparison of Means across Groups

Amandine Gamble
amandine.gamble@cornell.edu



House Keeping

Bonus Assignment

- R Cheat Sheet: Assignment now on Canvas
  - **Optional** (not mandatory), graded as bonus (up to + 1.5%)
  - Can be **any format** (R markdown html page, Illustrator poster, hand-written card…)
  - Can be **any language**
  - Should include **at least 10 commands, introduced in at least 2 different classes** (i.e., is updated as you learn)
  - Will be checked for plagiarism
  - Examples available on the Assignment page
  - Due at the end of the semester (May 5), but can be submitted earlier
  - **Main goal: have you reflect on what you have learned, and develop a useful resource**
  - **In particular: great way to reflect on AI-suggested commands**

Feedback on Previous Homework

How many observations are included in each category ?

```
table(male_18_data_hw2$Consumption)
#
```

Above Below Within 14 35 11 ...

- Good overall (=
- Easy way to get point: answering all the questions with explicit answers

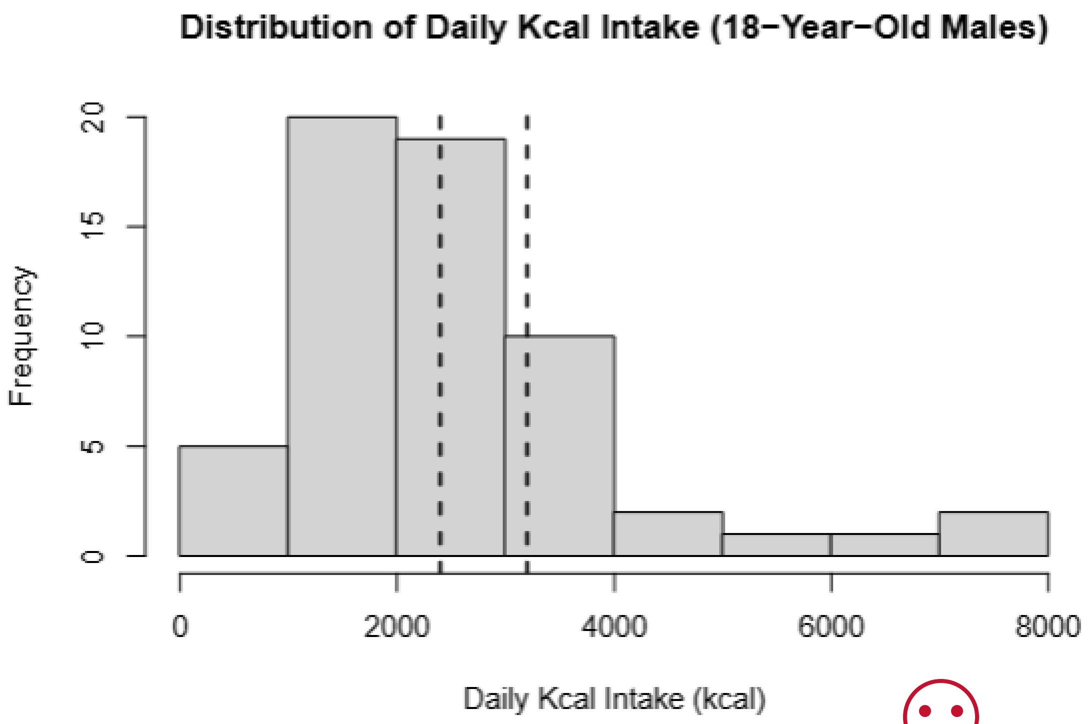


Figure 1: ...

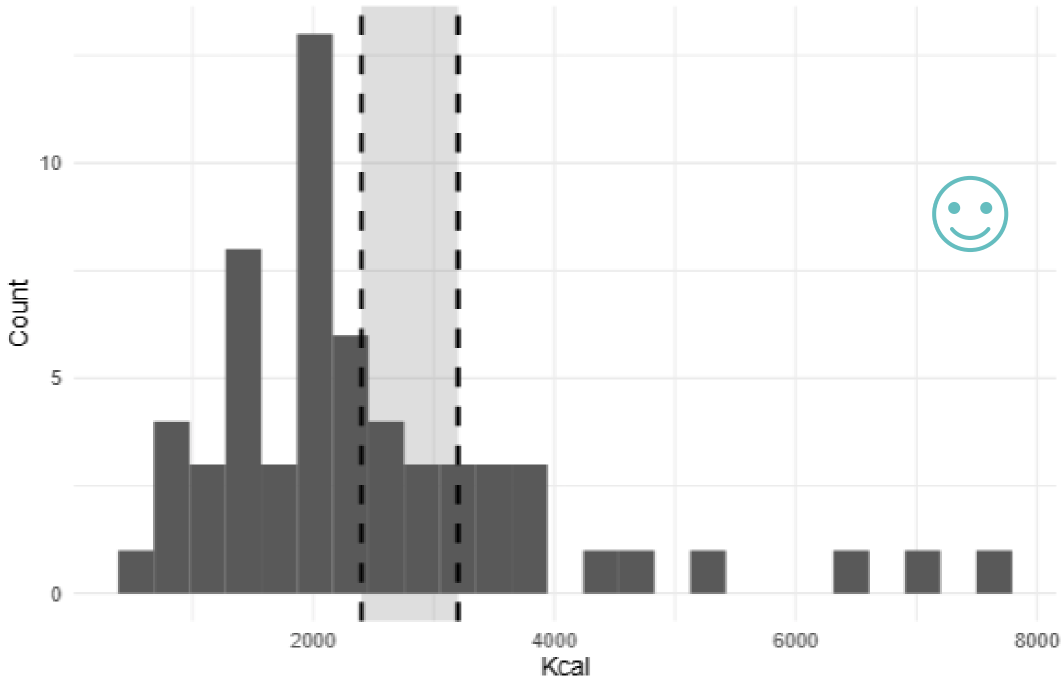
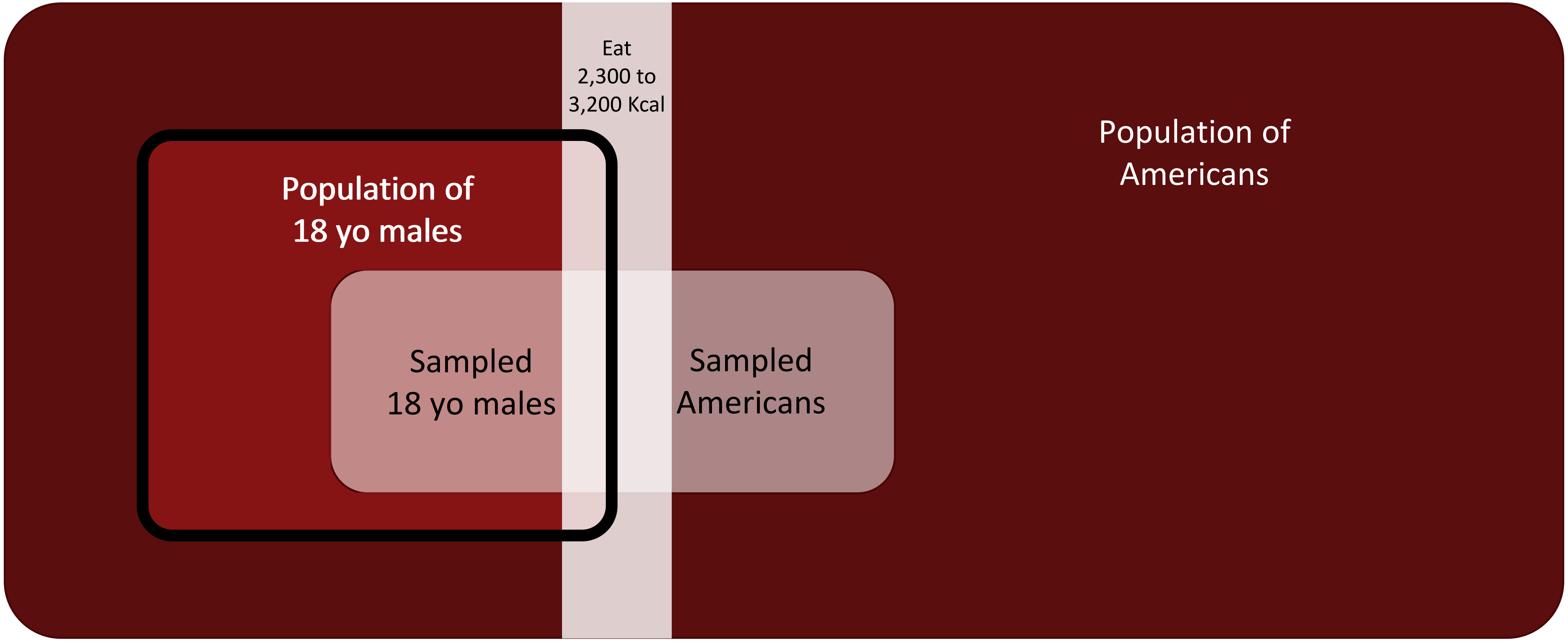


Figure 1: Distribution of Kcal intake among 18-year-old males with recommended range (2400–3200 Kcal).

Feedback on Previous Homework

- Good overall (=
- Easy way to get point: answering all the questions with explicit answers
- Probabilities to draw individuals within a sample, or a population…



Feedback on Previous Homework

- Good overall (=
- Easy way to get point: answering all the questions with explicit answers
- Probabilities to draw individuals within a sample, or a population…
- Coding…

```
data_adult_male <- data_male[data_male$AgeYear >= "18", ]
```

	Participant	Gender	AgeMonth	BMI	Diet	Kcal	KcalFromFat	CreamWithCoffee	Supplements	AgeYear
2	130401	Male	222	28.3	No	1788	612.00	No	No	18.500000
5	130411	Male	50	16.7	No	898	336.24	No	No	4.166667
6	130412	Male	88	21.6	No	1462	473.49	No	No	7.333333
8	130420	Male	223	34.4	No	756	239.13	No	No	18.583333
17	130479	Male	85	21.8	Yes	872	339.57	No	No	7.083333
21	130511	Male	64	16.4	No	1686	664.38	No	No	5.333333
39	130660	Male	62	15.8	No	1919	829.08	No	No	5.166667
40	130682	Male	110	16.7	No	2544	947.07	No	No	9.166667
46	130699	Male	68	13.6	No	1458	526.59	No	No	5.666667
54	130740	Male	49	14.7	No	1292	644.49	No	No	4.083333
55	130757	Male	119	14.6	No	1458	538.65	No	No	9.916667
61	130790	Male	70	15.5	No	1318	253.53	No	No	5.833333
67	130819	Male	52	16.2	No	2037	775.89	No	No	4.333333
70	130824	Male	97	16.9	No	2166	645.93	No	No	8.083333

```
> 4.166667 >= "18"
[1] TRUE
> 4 >= "18"
[1] TRUE
> 19 >= "18"
[1] TRUE
> 2 >= "18"
[1] TRUE
```

Feedback on Previous Homework

- Good overall (=
- Easy way to get point: answering all the questions with explicit answers
- Probabilities to draw individuals within a sample, or a population…
- Coding…

Amandine Gamble
Why does this return true in R 4 >= "18" even though 4 is a numerical value, and "18" a character string?

AI
This happens because of R's **automatic type coercion** (type conversion) rules. When R compares different data types, it converts them to a common type using a hierarchy.

R's Type Coercion Hierarchy:

sql — Copy code
```sql
logical → numeric → character
```

When comparing `4 >= "18"`, R converts both to the **most flexible type** (character), so the comparison becomes:

r — Copy code
```r
"4" >= "18"
```

Character Comparison Rules:

R compares strings **lexicographically** (like dictionary order), character by character from left to right:

- First characters: `"4"` vs `"1"`
- Since `"4" > "1"` in ASCII/Unicode order, `"4" >= "18"` returns `TRUE`

Check your output!!!

```
View()
head()
unique(vector)
…
```

```
> 4.166667 >= "18"
[1] TRUE
> 4 >= "18"
[1] TRUE
> 19 >= "18"
[1] TRUE
> 2 >= "18"
[1] TRUE
```

Hypothesis Testing

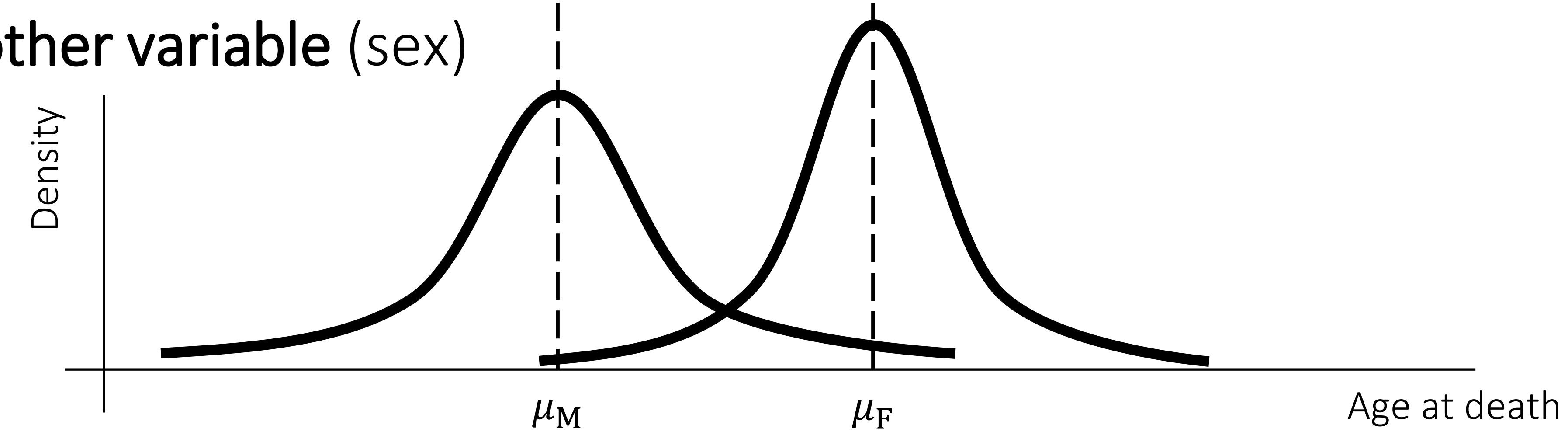
Testing Hypotheses…

- Hypothesis testing is the foundation of modern science
- **Assessing natural factors of variations**

  E.g., is sex associated with age of death?

  Hypothesis: age is associated with age of death

  If it is true, we expect the mean age of death for females to be different from the mean age of death for males, accounting for variations in age of death within each sex
- Notice that the **association between 2 variables** (sex and age of death) **translates into a differences in the values of one variable** (age) **across the values of the other variable** (sex)



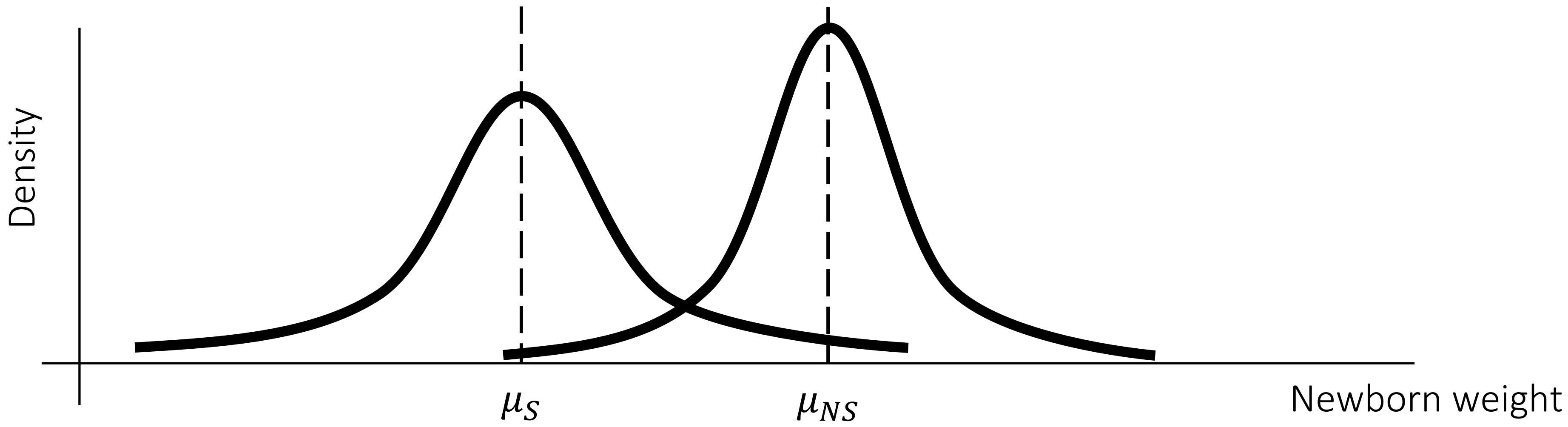
Testing Hypotheses…

- Hypothesis testing is the foundation of modern science
- **Identifying risk factors for diseases**

  E.g., is cannabis consumption associated with lower newborn weight?

  Hypothesis: cannabis consumption is associated with lower newborn weight

  If it is true, we expect newborns of mothers consuming cannabis to be lighter than newborns of mothers not consuming cannabis
- Notice that we can (but we don't have to) put an assumption on the **direction of the association**



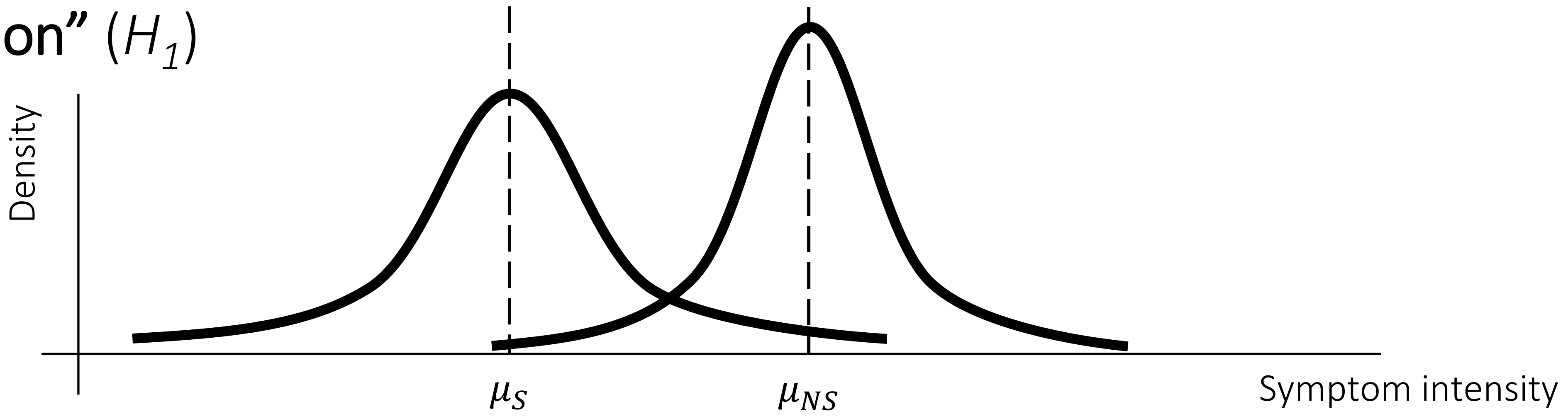
Testing Hypotheses…

- Hypothesis testing is the foundation of modern science
- **Testing the effects of treatments**

  E.g., is vitamin D supplementation associated with less severe COVID-19 symptoms?

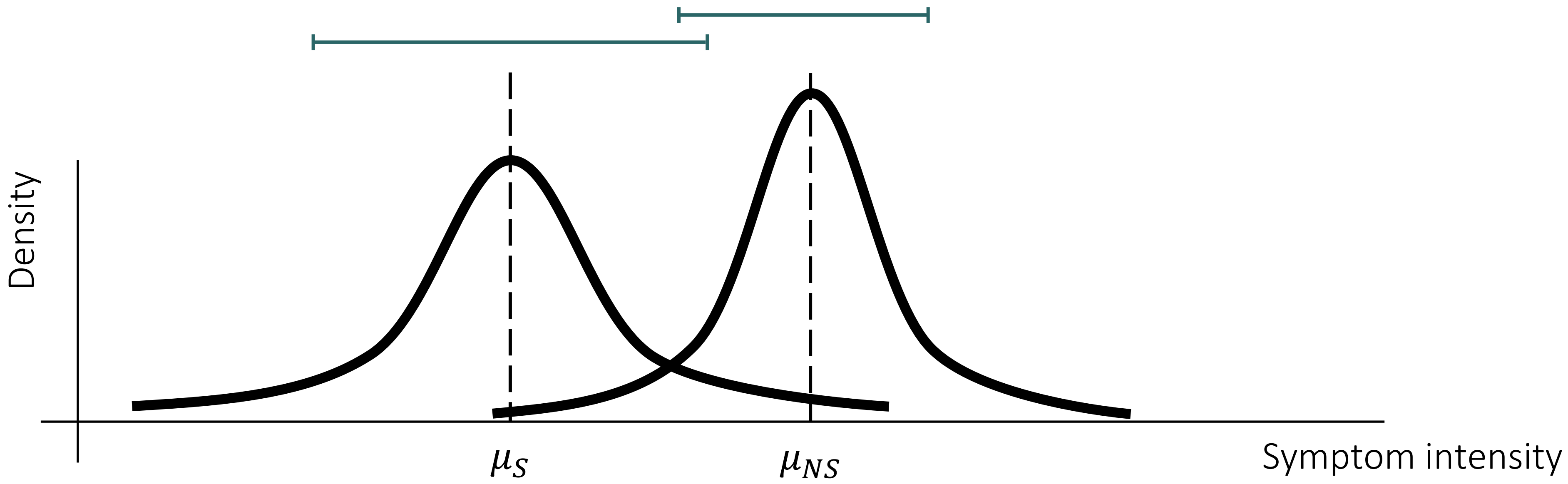
  Hypothesis: vitamin D supplementation is associated with less severe COVID-19 symptoms

  If it is true, we expect people supplemented with vitamin D to have less severe COVID-19 symptoms
- Notice that **the hypothesis is always formulated such as "there is an association"** (H_1)



Testing Hypotheses: the Concept

- Quantifying the **overlap** between the subpopulations of interest for the variable of interest, to assess whether or not the variable **significantly differs** across the subpopulations



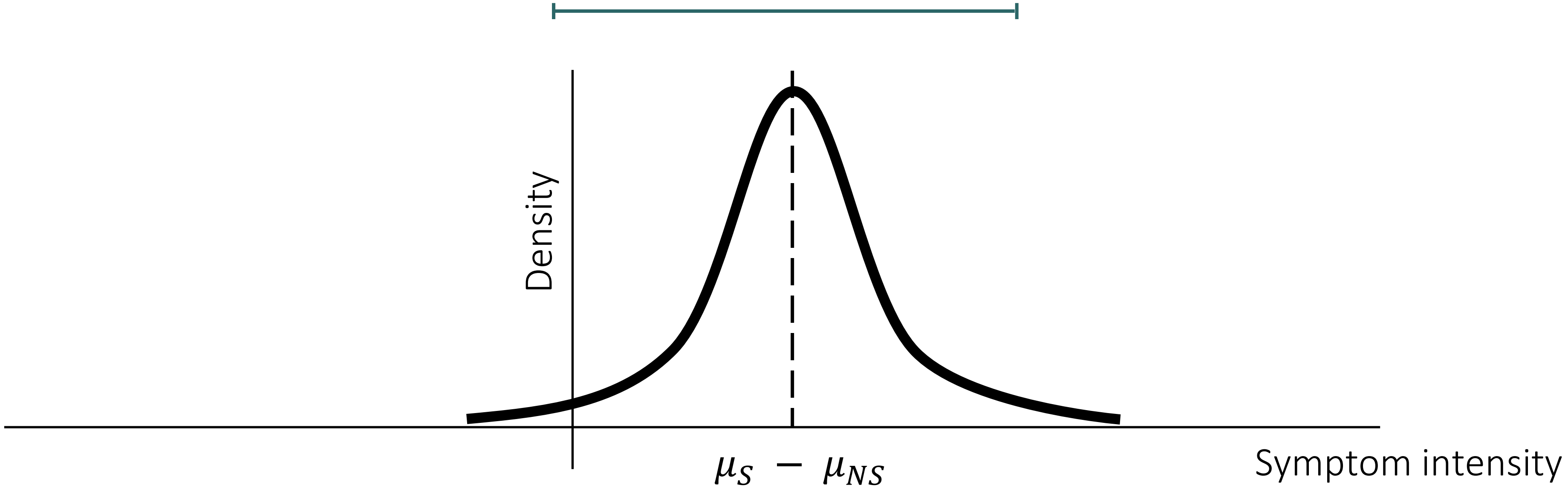
Testing Hypotheses Requires Hypotheses with Uncertainty

- Formulate a hypothesis, corresponding to the absence of association (H_0)
- Test it
- Reject it or not, with a given level of confidence (α)

Null hypothesis is…	True ($\mu_1 = \mu_2$)	False ($\mu_1 \neq \mu_2$)
Reject	Type I Error False positive Probability = α	Correct decision True positive (association) Probability = $1 - \beta$
Fail to reject	Correct decision True negative (no association) Probability = $1 - \alpha$	Type II Error False negative Probability = β

Testing Hypotheses: the Concept

- Quantifying the **overlap** between the subpopulations of interest for the variable of interest, to assess whether or not the variable **significantly differs** across the subpopulations
- Another way to do it: **quantifying the overlap with 0 of the difference of the variable between the subpopulations**



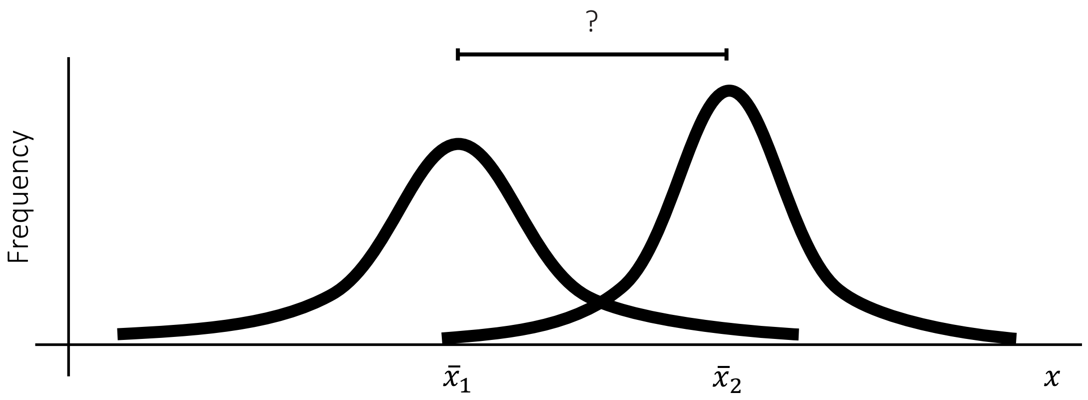
Testing Hypotheses Requires Hypotheses with Uncertainty

- Formulate a hypothesis, corresponding to the absence of association (H_0)
- Test it
- Reject it or not, with a given level of confidence (α)

Null hypothesis is…	True ($\mu_1 = \mu_2$) $\mu_1 - \mu_2 = 0$	False ($\mu_1 \neq \mu_2$) $\mu_1 - \mu_2 \neq 0$
Reject	Type I Error False positive Probability = α	Correct decision True positive (association) Probability = $1 - \beta$
Fail to reject	Correct decision True negative (no association) Probability = $1 - \alpha$	Type II Error False negative Probability = β

Comparison of Means

The basics: the confidence interval method



Confidence Intervals for Differences

Mean

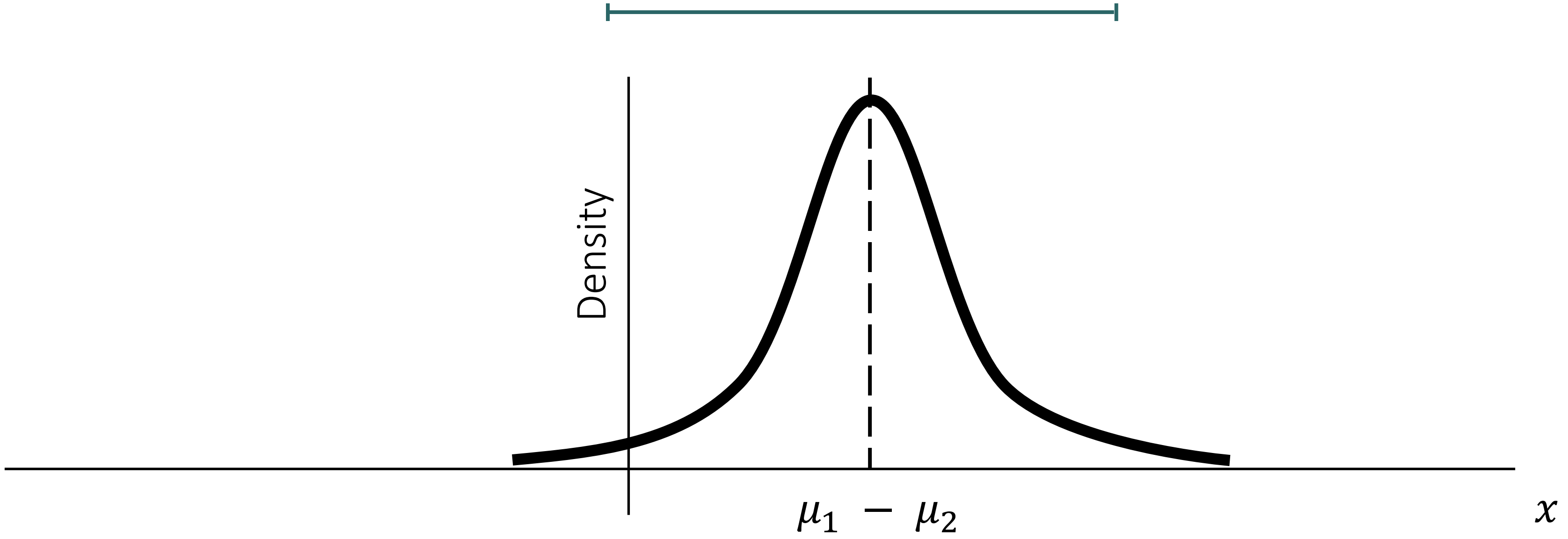
$$\text{CI} = (\bar{x}_1 - \bar{x}_2) \pm z\sqrt{\frac{{\sigma_{s_1}}^2}{n_1} + \frac{{\sigma_{s_2}}^2}{n_2}}$$

If the variance is the same

$$\text{CI} = (\bar{x}_1 - \bar{x}_2) \pm z\sigma_s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Proportion

$$\text{CI} = (\bar{p}_1 - \bar{p}_2) \pm z\sqrt{\frac{\bar{p}_1(1-\bar{p}_1)}{n_1} + \frac{\bar{p}_2(1-\bar{p}_2)}{n_2}}$$



Confidence Intervals for Differences

Mean

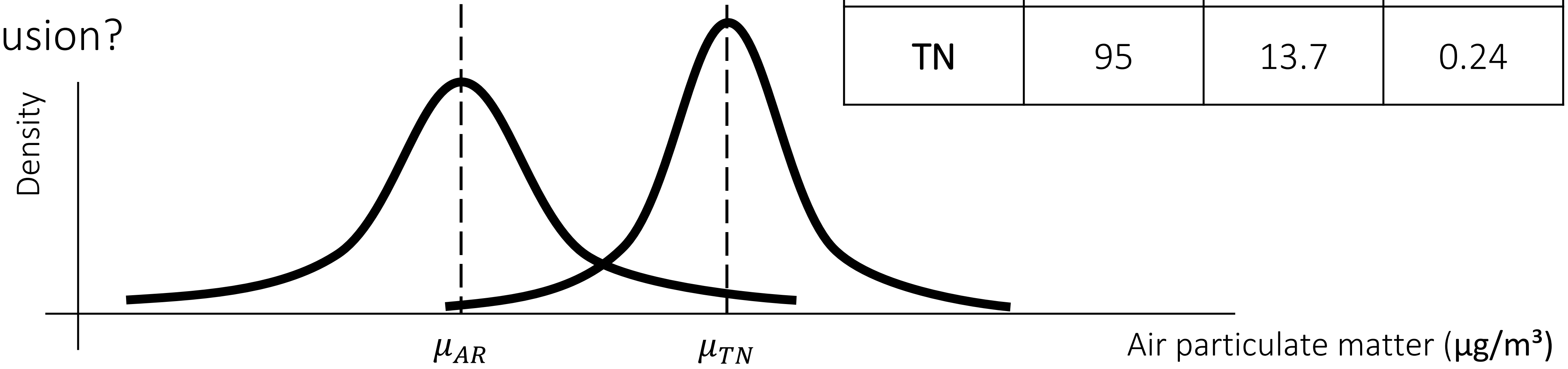
$$\text{CI} = (\bar{x}_1 - \bar{x}_2) \pm z\sqrt{\frac{{\sigma_{s_1}}^2}{n_1} + \frac{{\sigma_{s_2}}^2}{n_2}}$$

$$\text{CI} = -2.16, -1.84$$

We reject the hypothesis of no difference with confidence level 95%

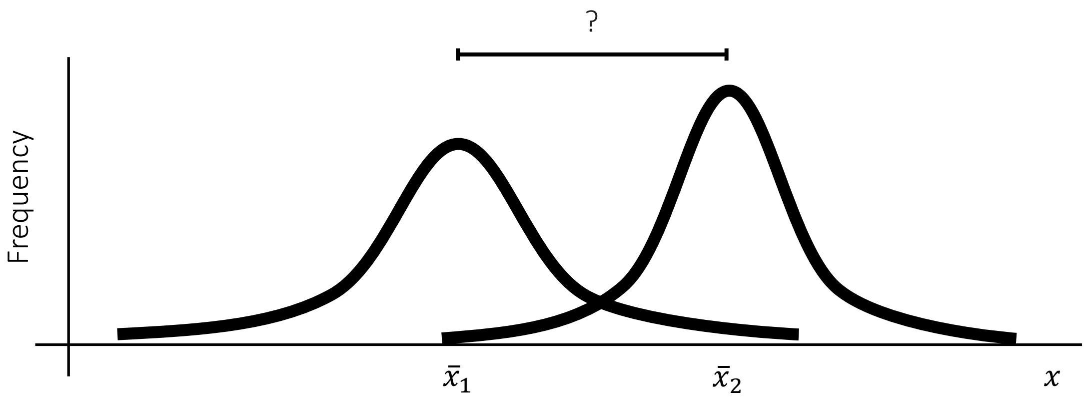
- Compare the average pollution by air particulate matter between Arkansas and Tennessee
  - H_0 and H_1?
  - CI of the difference?
  - Conclusion?

Data	n	$\bar{x}$	σ_s
AR	75	11.7	0.32
TN	95	13.7	0.24

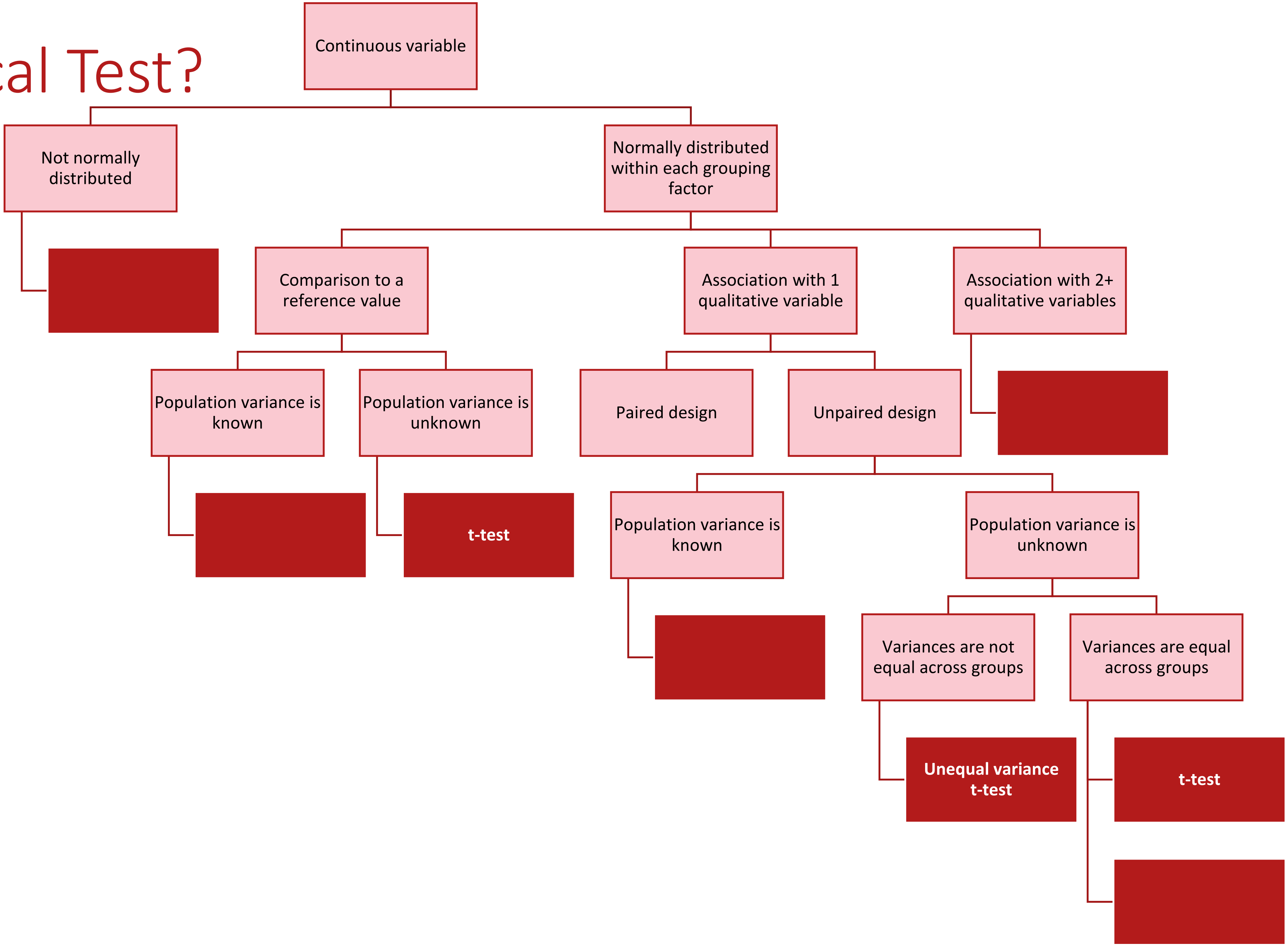


Comparison of Means

The common one: t-test



Which Statistical Test?

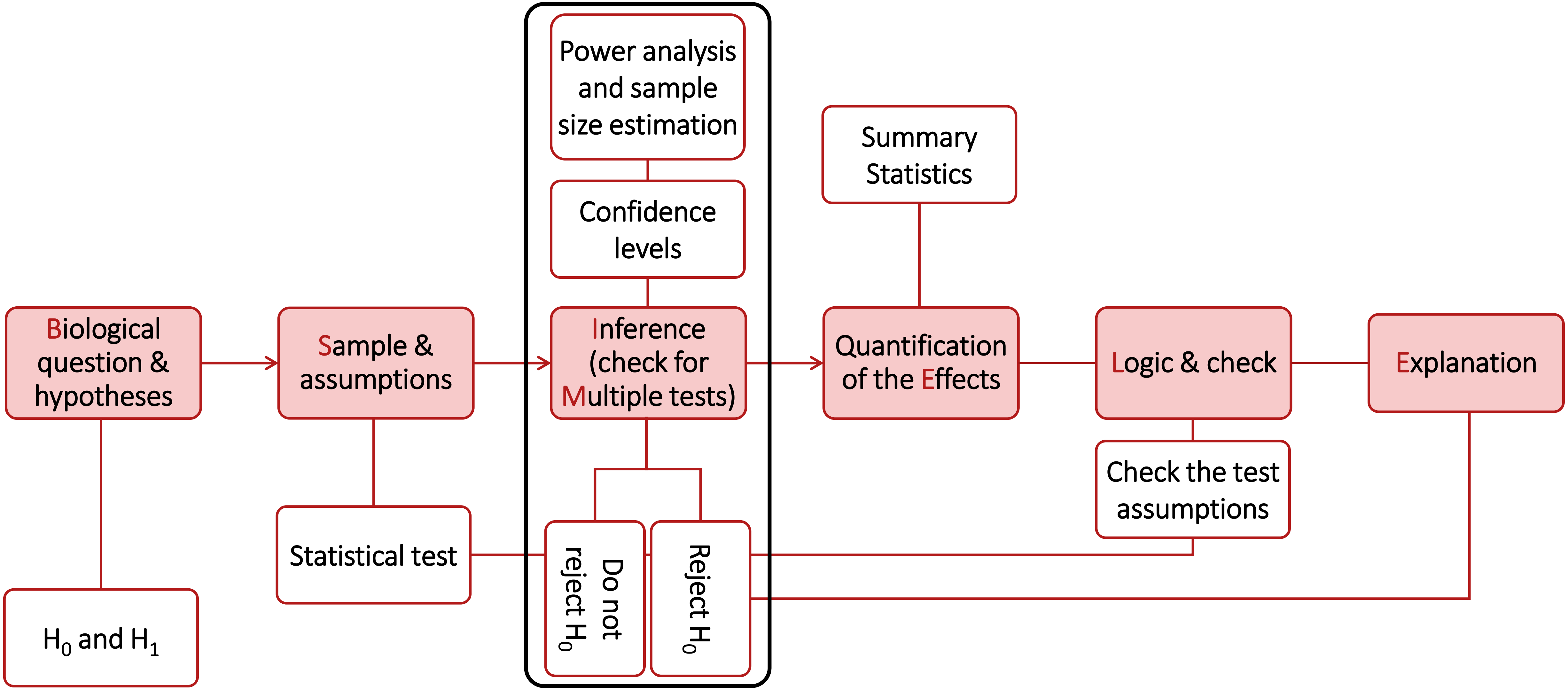


Hypothesis Testing Considerations

B-SIMPLE

- **Biological question and research hypothesis** What am I actually trying to answer?
- **Sample and assumptions** Which approach can I reasonably use?
- **Inference power and significance** Can we, and do we, detect an association?
- **Multiple comparisons and adjustments** Do the thresholds need to be adjusted for multiple testing?
- **Practical effect** What is the magnitude of the association?
- **Logic and check** Does the output make sense? Are the test assumptions verified?
- **Explanation** How do I communicate the results to my audience?

Hypothesis Testing Workflow

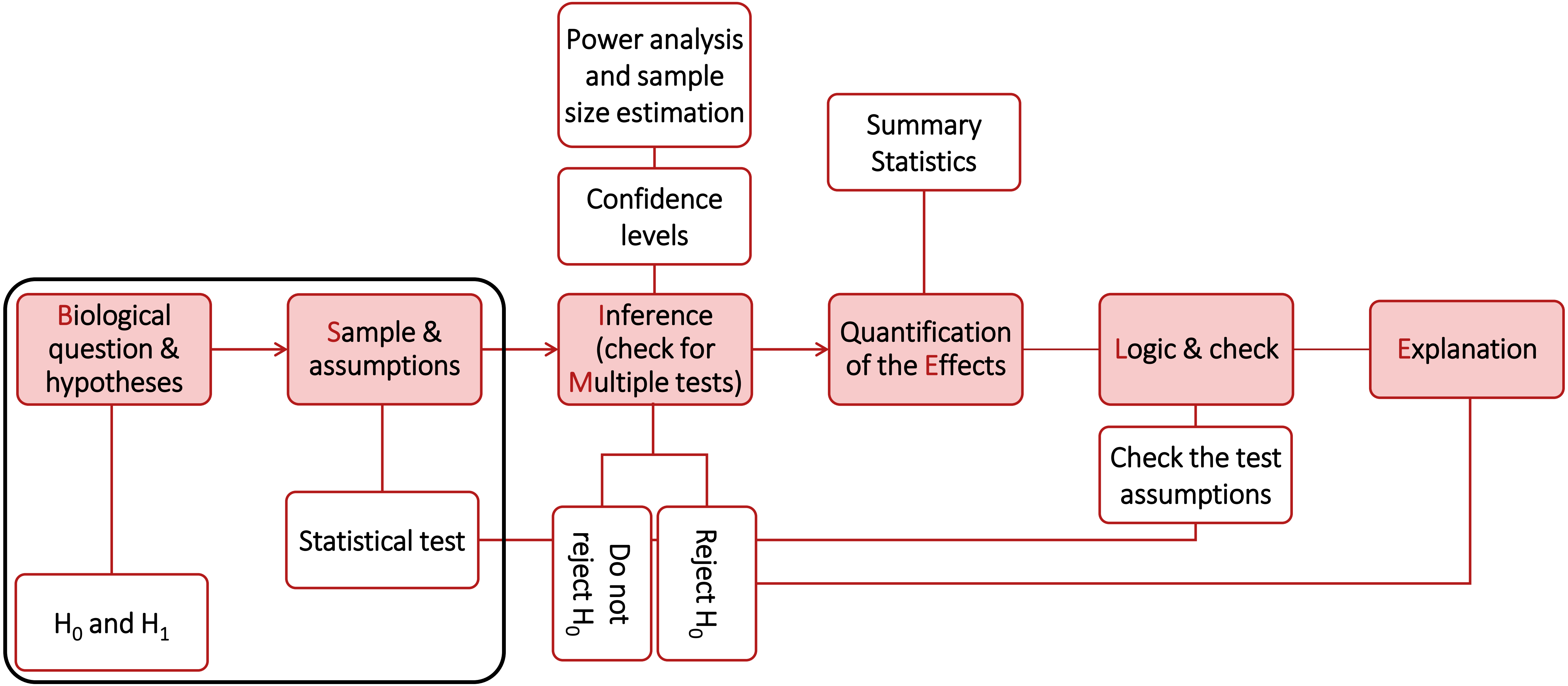


Confidence, Significance, and Power: Same Concept for All Tests

- Null hypothesis (H_0): absence of association between variables, i.e., absence of difference between groups ($\mu_1 = \mu_2$)
- Alternative hypothesis (H_1): association between variables, i.e., difference between groups ($\mu_1 \neq \mu_2$)

Null hypothesis (H_0) is…	True ($\mu_1 = \mu_2$)	False ($\mu_1 \neq \mu_2$)
Reject	Type I Error False positive **Probability of Type I Error =** $\boldsymbol{\alpha}$ **= significance**	Correct decision True positive (association) **Probability of no Type II Error =** $\mathbf{1} - \boldsymbol{\beta}$ **= power**
Fail to reject	Correct decision True negative (no association) **Probability of no Type I Error =** $\mathbf{1} - \boldsymbol{\alpha}$ **= confidence**	Type II Error False negative **Probability of Type II Error =** $\boldsymbol{\beta}$

Hypothesis Testing Workflow



Statistical Tests: the T-test, When?

- Comparing 2 means or 2 proportions together

  ⇒ Testing the association between a binary variable and a continuous variable

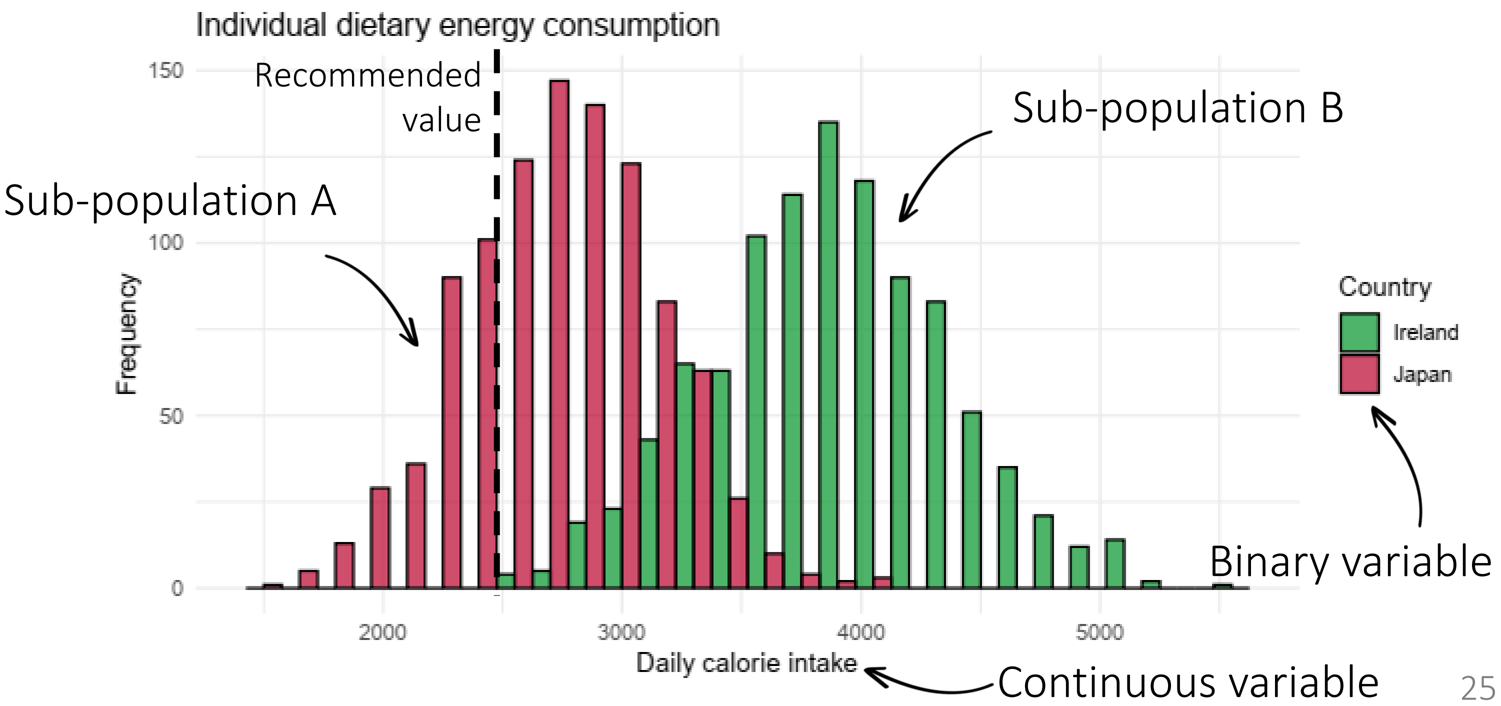
  $\mu_1 = \mu_2$?

  Two-sample t-test

- 1 mean or 1 proportion to a reference value:

  $\mu_1 = \mu_0$?

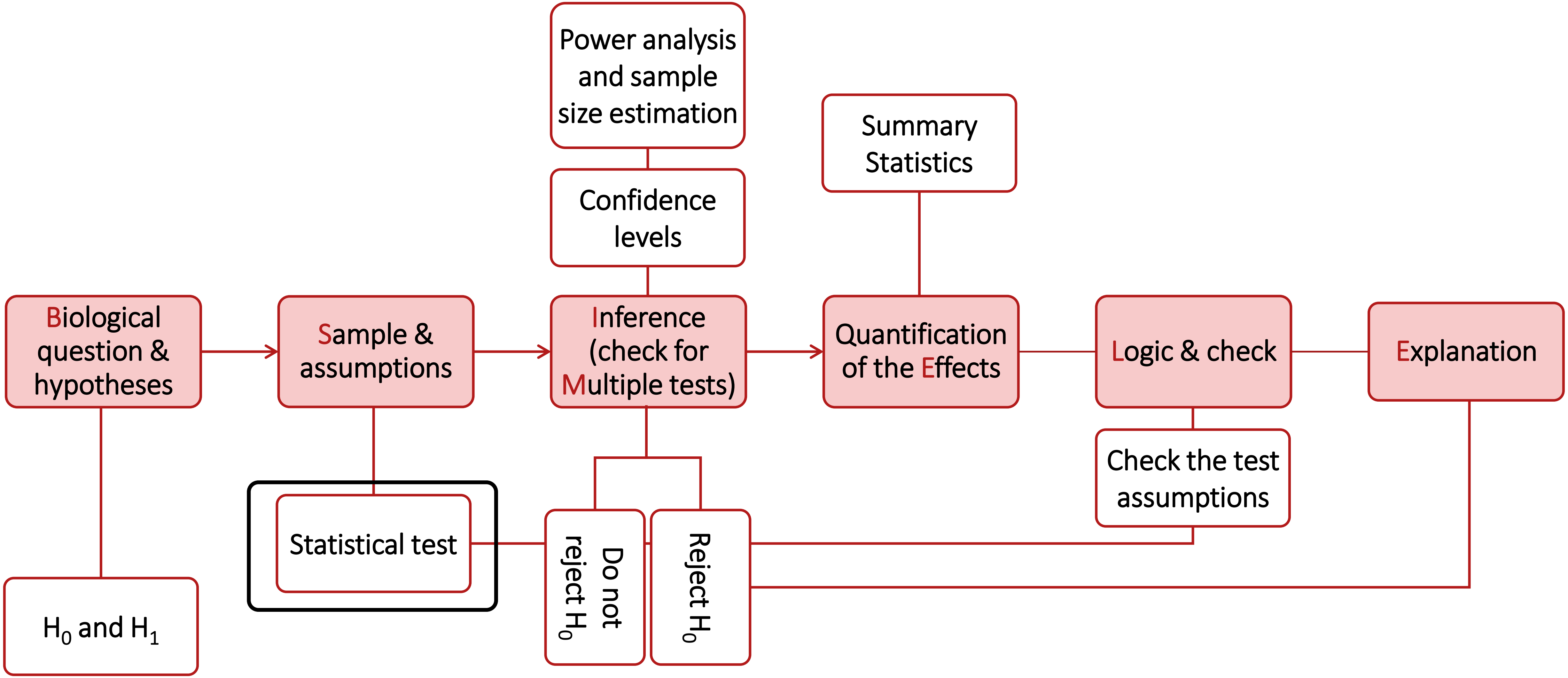
  One-sample t-test



Statistical Tests: the T-test, When?

- Comparing 2 means or 2 proportions together
  Or 1 mean or 1 proportion to a reference value
- Assumptions on the data
  - Data is **normally distributed** within each group
  - The **variance of each population…**
    - **is known 😬 → z-test**
    - **is unknown → t-test**
  - Observations from each population…
    - are **independent** of one another **→ unpaired t-test**
    - are paired → paired t-test (next lecture)
  - Variances…
    - are equal across populations (homoscedasticity) → t-statistics formular for equal variances
    - are not equal across populations → t-statistics formular for non-equal variances

Hypothesis Testing Workflow



One-sample T-test

Statistical Tests: the T-test, How?

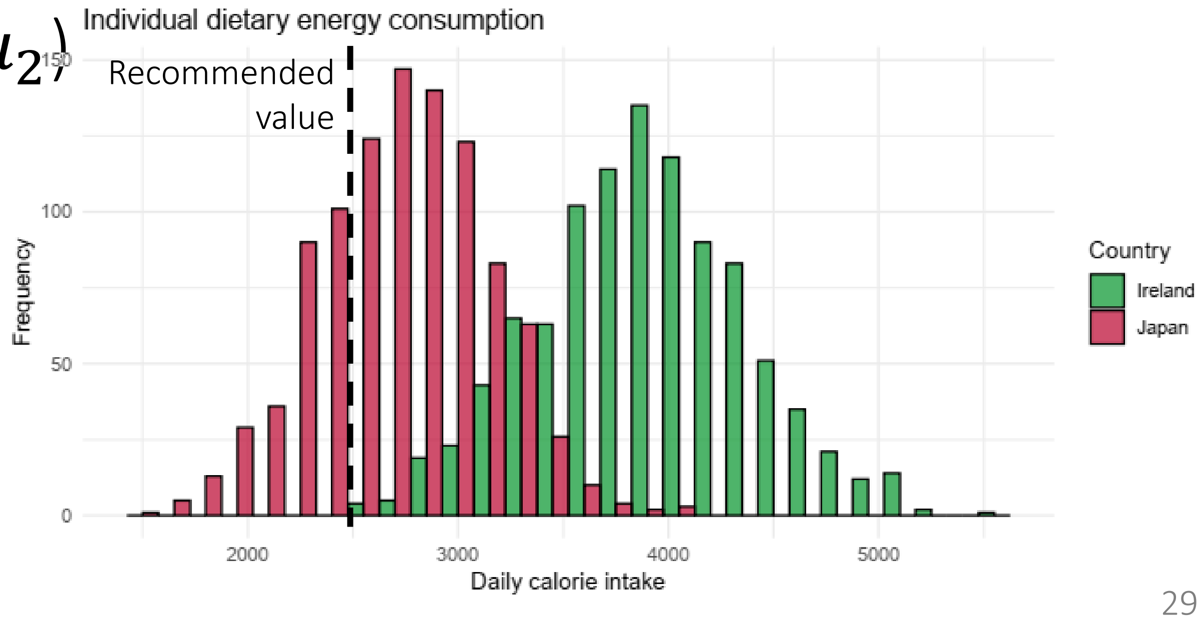
- One-sample: 1 mean to a reference value (μ and μ_0)

$$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}, df = n - 1$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ the sample standard deviation and n the sample size

With $\bar{x}$ close to μ_0, hence z close to 0 if H_0 is true

- Two-sample: 2 means together (μ_1 and μ_2)



Statistical Tests: the T-test, How?

- One-sample: 1 mean to a reference value (μ and μ_0)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, df = n - 1$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ <u>**the sample standard deviation**</u> and n the sample size

With $\bar{x}$ close to μ_0, hence z close to 0 if H_0 is true

Observed

Would be observed under H_0

The formula depends on whether the sample variances are equal or not, and is a mouthful… Run the test in R instead =)

- Two-sample: 2 means together (μ_1 and μ_2), for equal variance

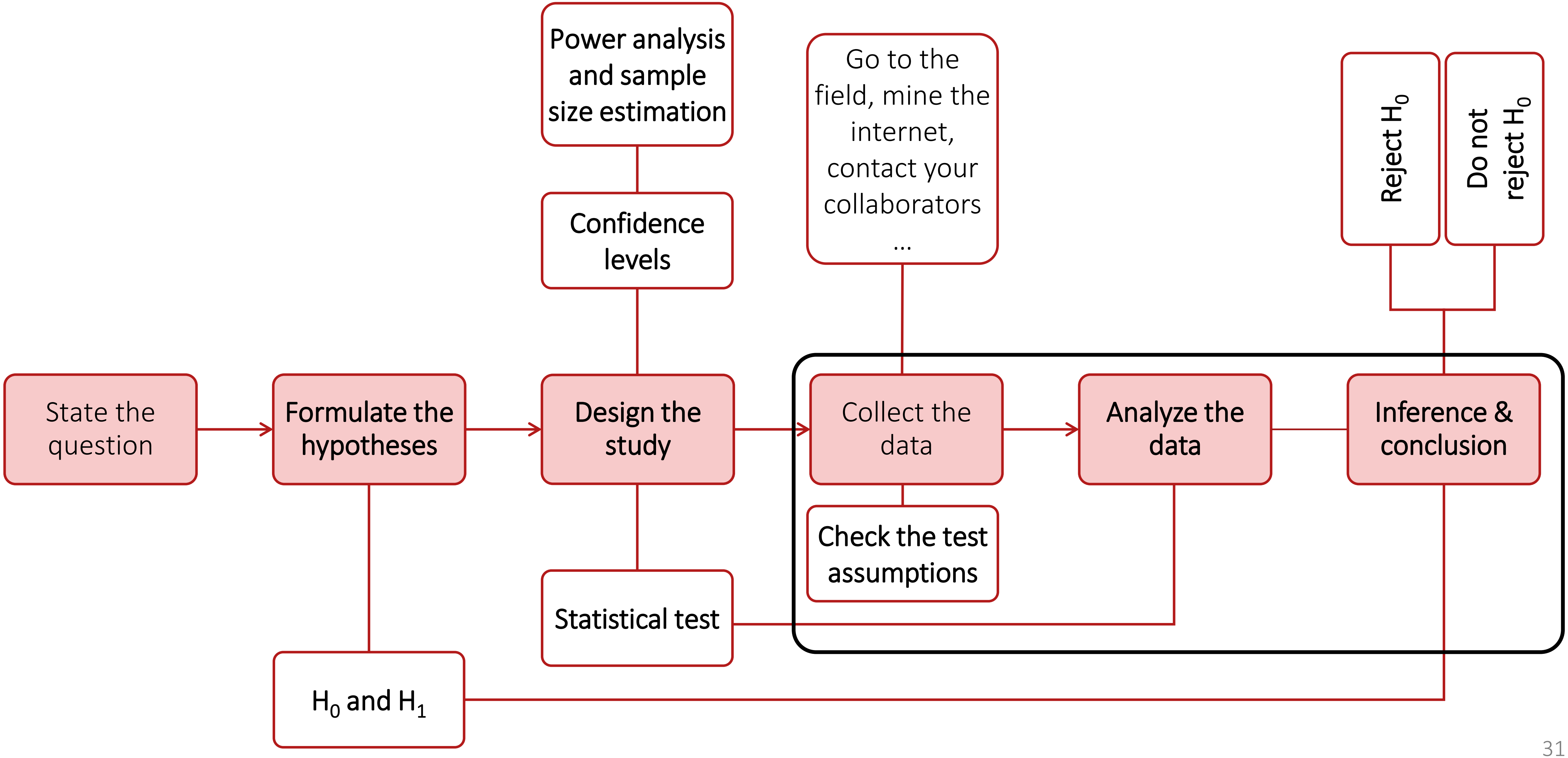
Compared to a known distribution

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2 \frac{1}{n_1} - \frac{1}{n_2}}}, df = n_1 + n_2 - 2$$

Standardized by variation

With $\mu_1 - \mu_2 = 0$ under H_0, and $\bar{x}_1 - \bar{x}_2$close to 0, hence z close to 0 if H_0 is true

The hypothesis testing workflow



Statistical Tests: the T-test in Practice… Assumptions

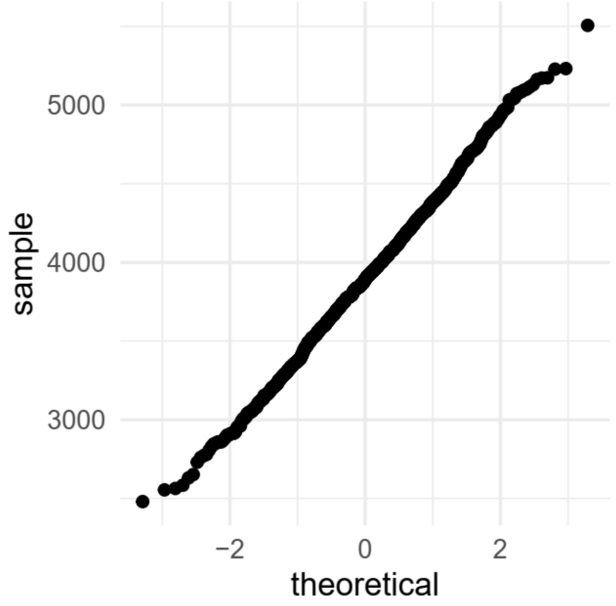
- One-sample: 1 mean to a reference value (μ and μ_0)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, df = n - 1$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ the sample standard deviation and n the sample size

With $\bar{x}$ close to μ_0, hence z close to $\mathbf{0}$ if H_0 is true

- Do people in Ireland, on average, consume more than the recommended daily calorie intake (2,500 calories)?
- Assumptions
  - Data are normally distributed within each group
    - Visual examination of the histogram
    - Visual examination of the QQ-plot
    - Shapiro-Wilk test for comparison of the data with a normal distribution (returns a p-value)



Statistical Tests: the T-test in Practice… Calculation

- One-sample: 1 mean to a reference value (μ and μ_0)

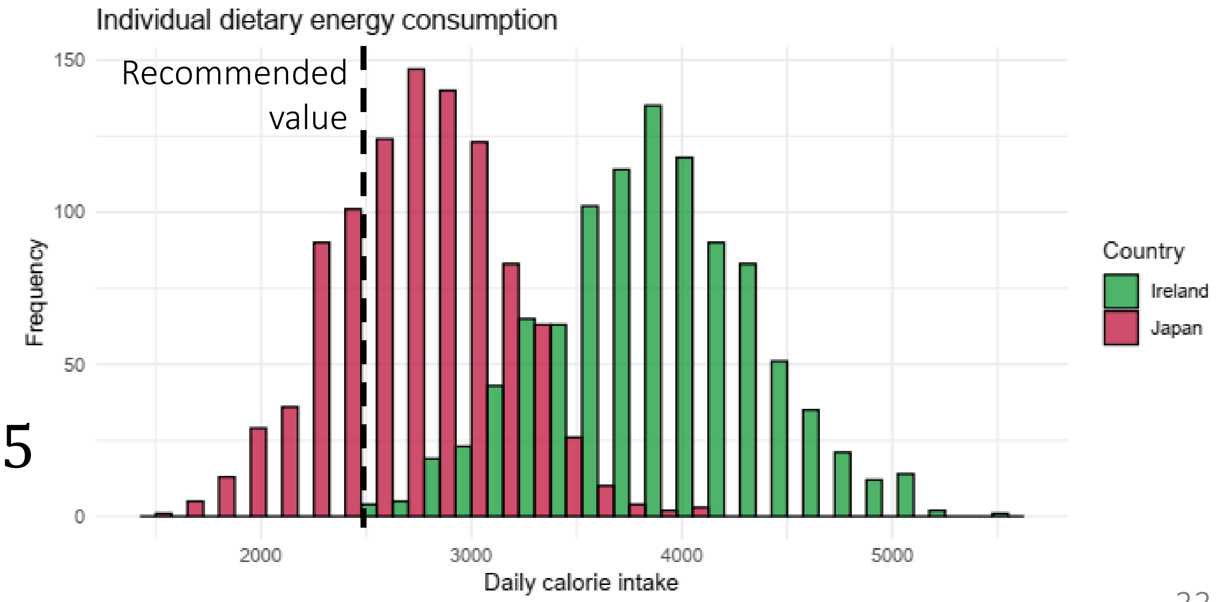
$$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}, df = n - 1$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ the sample standard deviation and n the sample size

With $\bar{x}$ close to μ_0, hence z close to 0 if H_0 is true

- Do people in Ireland, on average, consume more than the recommended daily calorie intake (2,500 calories)?

Country	Sample size	Mean	Standard deviation
Ireland	1,000	3,893	495



$$t = \frac{3{,}893 - 2{,}500}{495 / \sqrt{1000}} = 88.84294 \quad \begin{matrix} > 1.96 \\ < -1.96 \end{matrix} \qquad p < 0.05$$

$$df = 1{,}000 - 1 = 999 \qquad \approx 1{,}000$$

t-table

Statistical Tests: the T-test in Practice...

We reject H_0

Ireland average daily calorie intake is statistically different from the recommended value, with a level of confidence of 95%

- One-sample: 1 mean to a reference value (μ and μ_0)

$$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}, df = n-1$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ **the sample standard deviation** and n the sample size

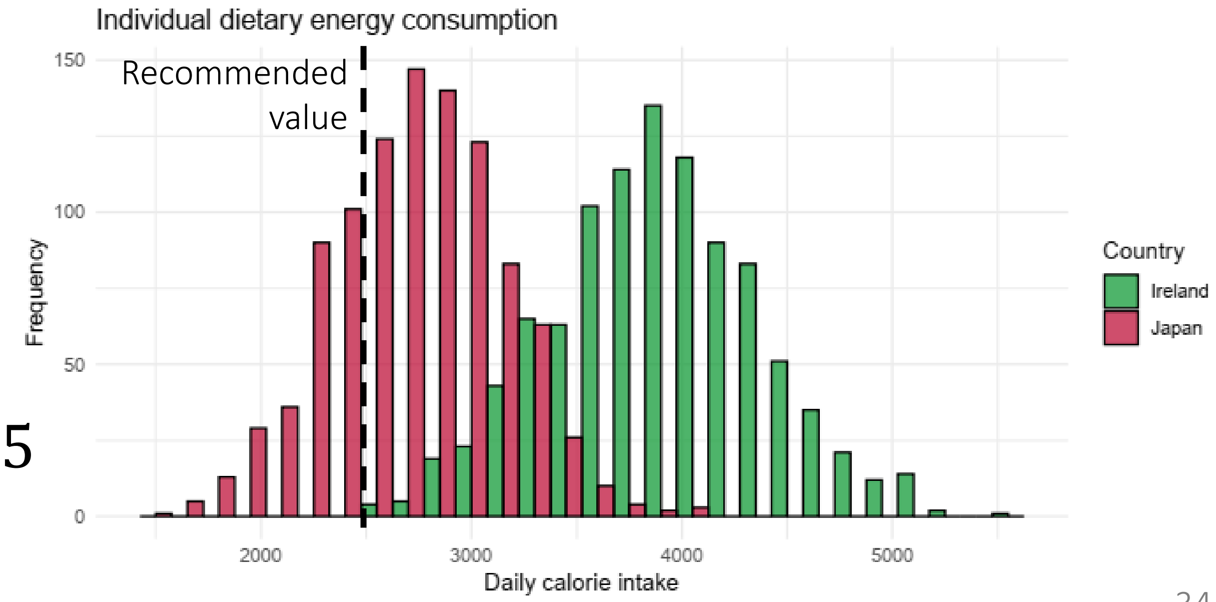
With $\bar{x}$ close to μ_0, hence z close to 0 if H_0 is true

- Do people in Ireland, on average, consume more than the recommended daily calorie intake (2,500 calories)?

Country	Sample size	Mean	Standard deviation
Ireland	1,000	3,893	495

$$t = \frac{3{,}893 - 2{,}500}{495/\sqrt{1000}} = 88.84294 \quad \begin{matrix} > 1.96 \\ < -1.96 \end{matrix} \qquad p < 0.05$$

$$df = 1{,}000 - 1 = 999 \quad \approx 1{,}000$$



t-table

Statistical Tests: the T-test in Practice…

We reject H_0

Japan average daily calorie intake is statistically different from the recommended value, with a level of confidence of 95%

- One-sample: 1 mean to a reference value (μ and μ_0)

$$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}, df = n - 1$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ **the sample standard deviation** and n the sample size

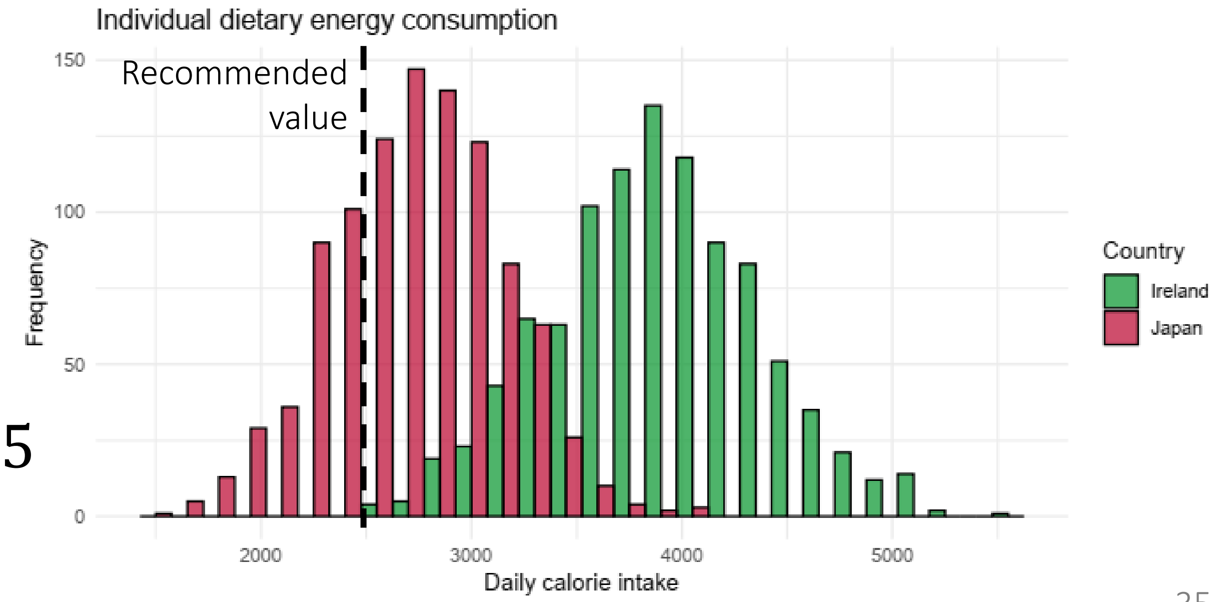
With $\bar{x}$ close to μ_0, hence z close to 0 if H_0 is true

- Do people in Japan, on average, consume more than the recommended daily calorie intake (2,500 calories)?

Country	Sample size	Mean	Standard deviation
Japan	1,000	2,722	404

$$t = \frac{2{,}722 - 2{,}500}{404/\sqrt{1000}} = 17.38 \quad \begin{matrix} > 1.96 \\ < -1.96 \end{matrix} \quad p < 0.05$$

$$df = 1{,}000 - 1 = 999 \quad \approx 1{,}000$$



t-table

Statistical Tests: the T-test in Practice…

We do not reject H_0

Japan average daily calorie intake is not statistically different from the recommended value

We lack **power**…

- One-sample: 1 mean to a reference value (μ and μ_0)

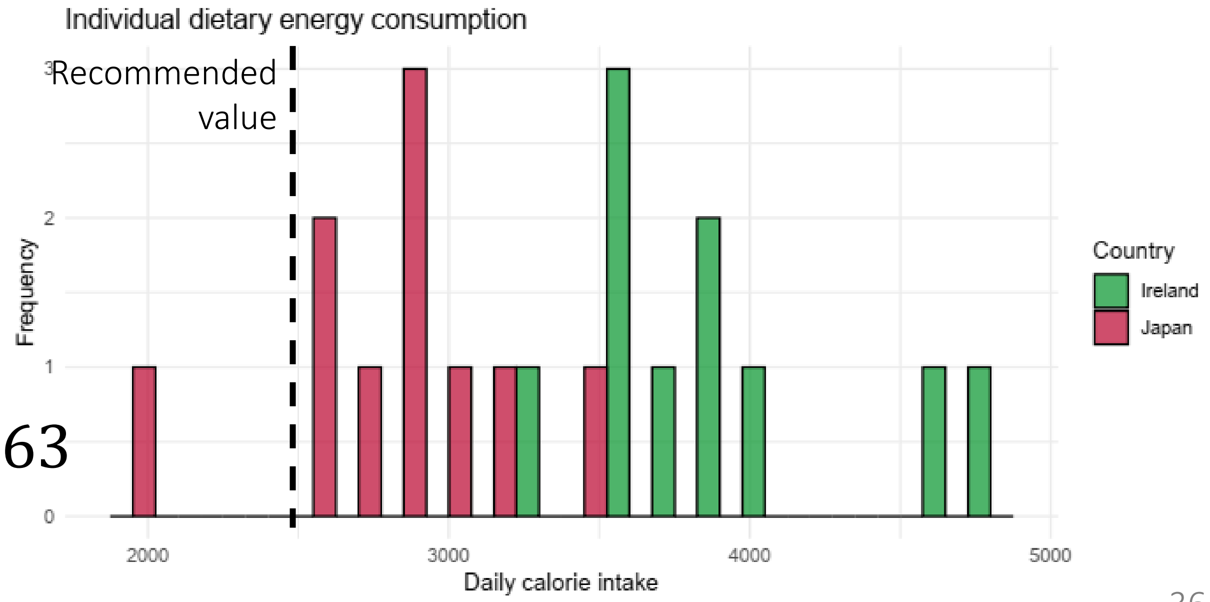
$$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}, df = n-1$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ **the sample standard deviation** and n the sample size

With $\bar{x}$ close to μ_0, hence z close to 0 if H_0 is true

- Do people in Japan, on average, consume more than the recommended daily calorie intake (2,500 calories)?

Country	Sample size	Mean	Standard deviation
Japan	10	2,788	415



$$t = \frac{2{,}788 - 2{,}500}{415\ /\sqrt{30}} = 2.1968 \quad \begin{matrix} \ldots < 2.262 \\ > -2.262 \end{matrix}$$

$$df = 10 - 1 = 9 \quad < 1{,}000$$

$p = 0.05563$ (from R)

[t-table]

Statistical Tests: the T-test in Practice…

We do not reject H_0

Japan average daily calorie intake is not statistically different from the recommended value

We lack **power**…

- One-sample: 1 mean to a reference value (μ and μ_0)

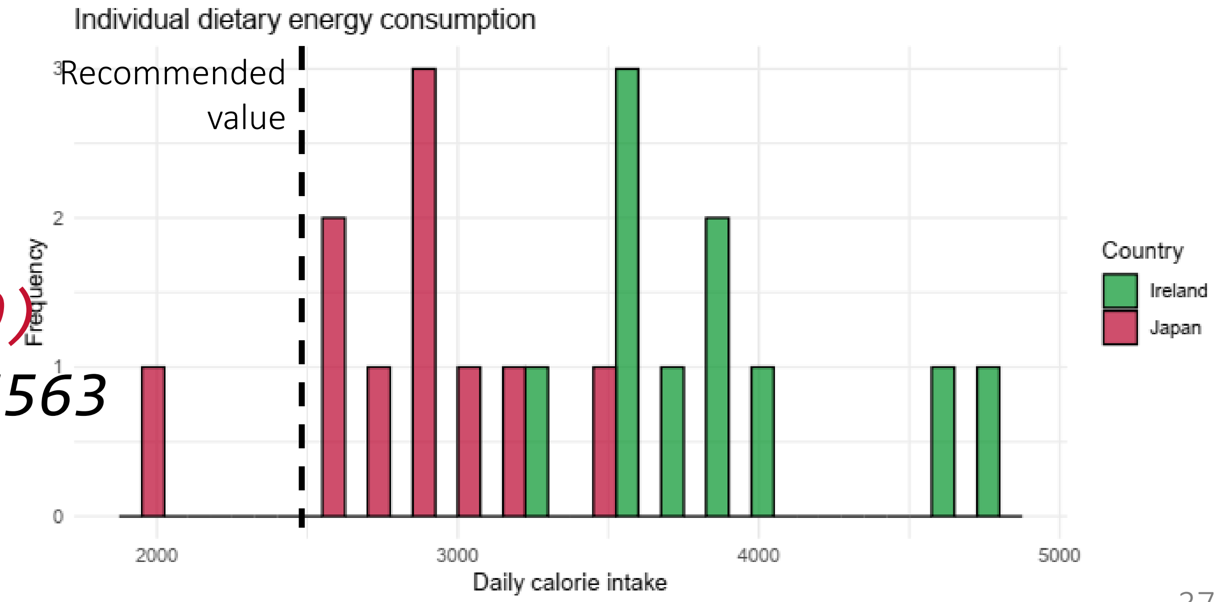
$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, df = n - 1$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ **the sample standard deviation** and n the sample size

With $\bar{x}$ close to μ_0, hence z close to 0 if H_0 is true

- Do people in Japan, on average, consume more than the recommended daily calorie intake (2,500 calories)?
- In R

```
t.test(data[data$country=="Japan",
       "calorie_intake"], mu = 2500)
t = 2.1968, df = 9, p-value = 0.05563
```



Two-sample T-test

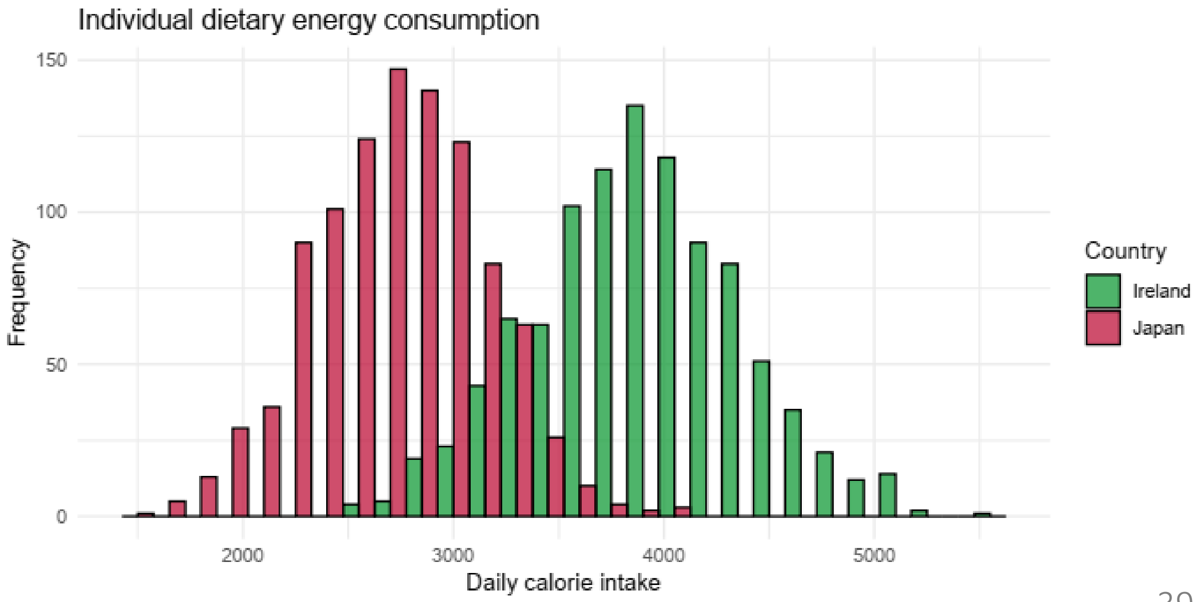
Statistical Tests: the T-test in Practice...

- Two-sample: 2 means together (μ_1 and μ_2)

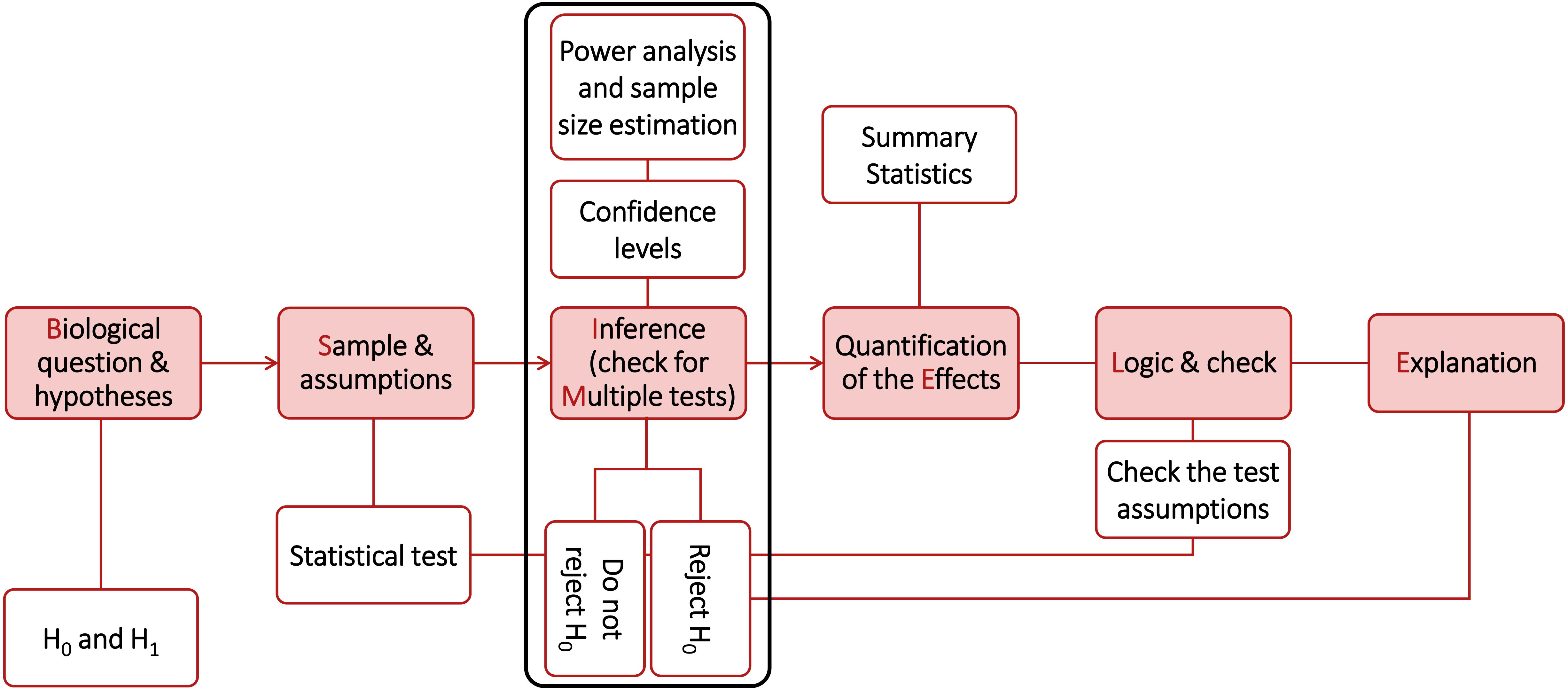
$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2 \frac{1}{n_1} - \frac{1}{n_2}}}, df = n_1 + n_2 - 2$$

With $\mu_1 - \mu_2 = 0$ under H_0, and $\bar{x}_1 - \bar{x}_2$close to 0, hence z close to 0 if H_0 is true

- Is there a difference between Japan and Ireland average daily calorie intake?
- Assumptions
  - Data are normally distributed within each group
  - Samples from each population are independent of one another
  - **Homoscedasticity (variances are equal across populations)** → Visual examination



Hypothesis Testing Workflow



Statistical Tests: the T-test, How?

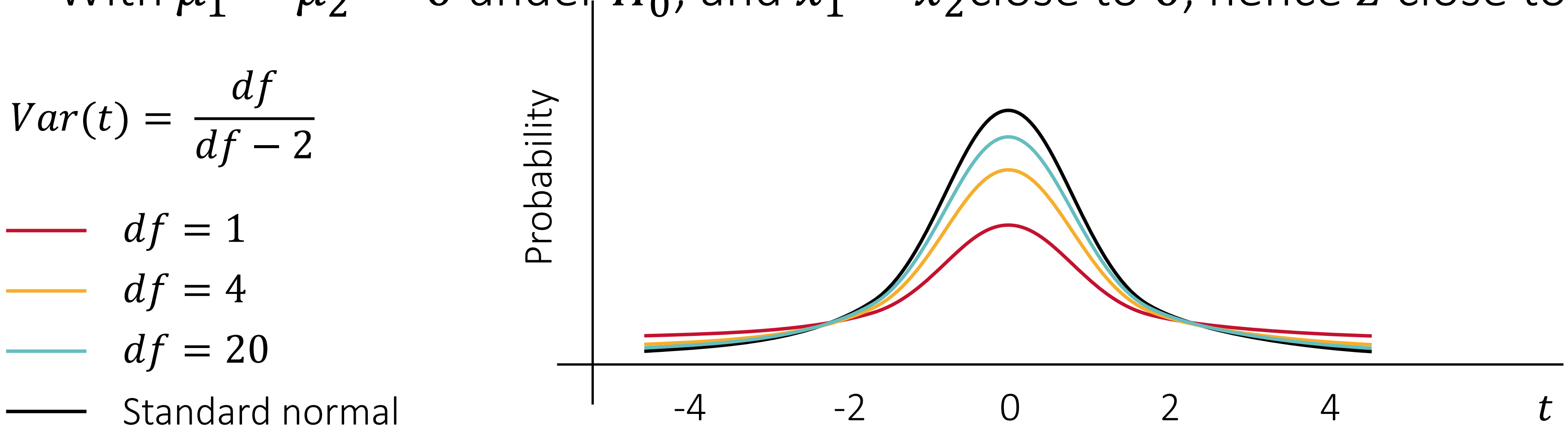
- Two-sample: 2 means together (μ_1 and μ_2)

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2 \frac{1}{n_1} - \frac{1}{n_2}}}, df = n_1 + n_2 - 2$$

With $\bar{x}$ the sample mean, μ_0 the reference value, $\boldsymbol{s}$ **<u>the sample standard deviation</u>** and n the sample size

With $\mu_1 - \mu_2 = 0$ under H_0, and $\bar{x}_1 - \bar{x}_2$ close to 0, hence z close to 0 if H_0 is true

$$Var(t) = \frac{df}{df - 2}$$



$t \to z$ when $df \to \infty$

Statistical Tests: the T-test, How?

- Two-sample: 2 means together (μ_1 and μ_2)

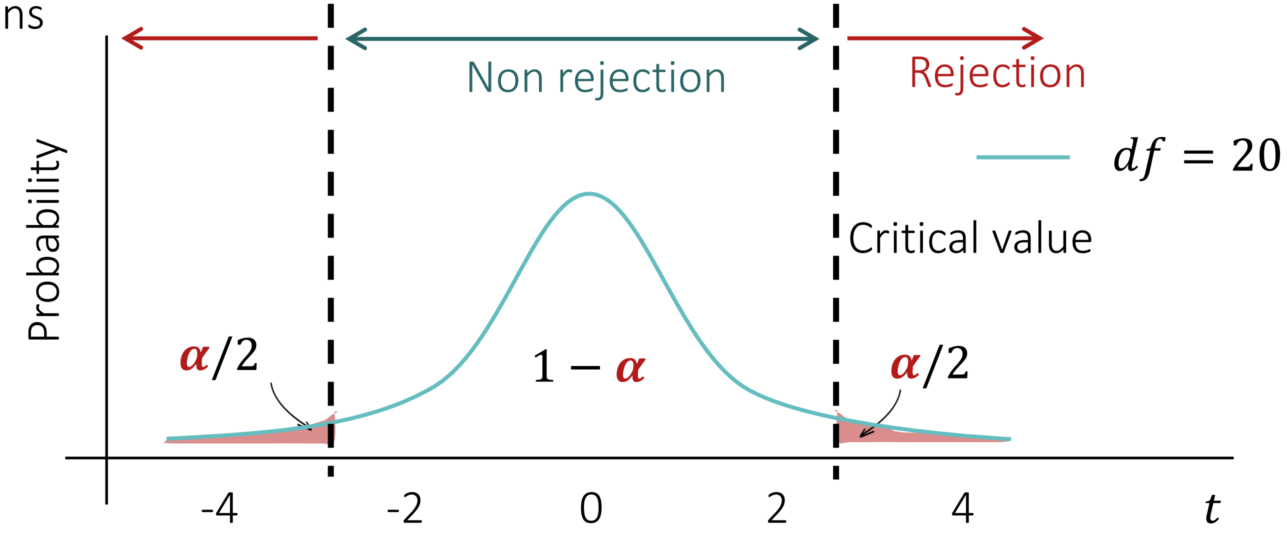
$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2 \frac{1}{n_1} - \frac{1}{n_2}}}, df = n_1 + n_2 - 2$$

With $\bar{x}$ the sample mean, μ_0 the reference value, ***s*** **the sample standard deviation** and n the sample size

With $\mu_1 - \mu_2 = 0$ under H_0, and $\bar{x}_1 - \bar{x}_2$close to 0, hence z close to 0 if H_0 is true

Known distributions

t-table



Statistical Tests: the T-test, How?

In practice, set α, provide the data, and R will compute the p-value

t Table

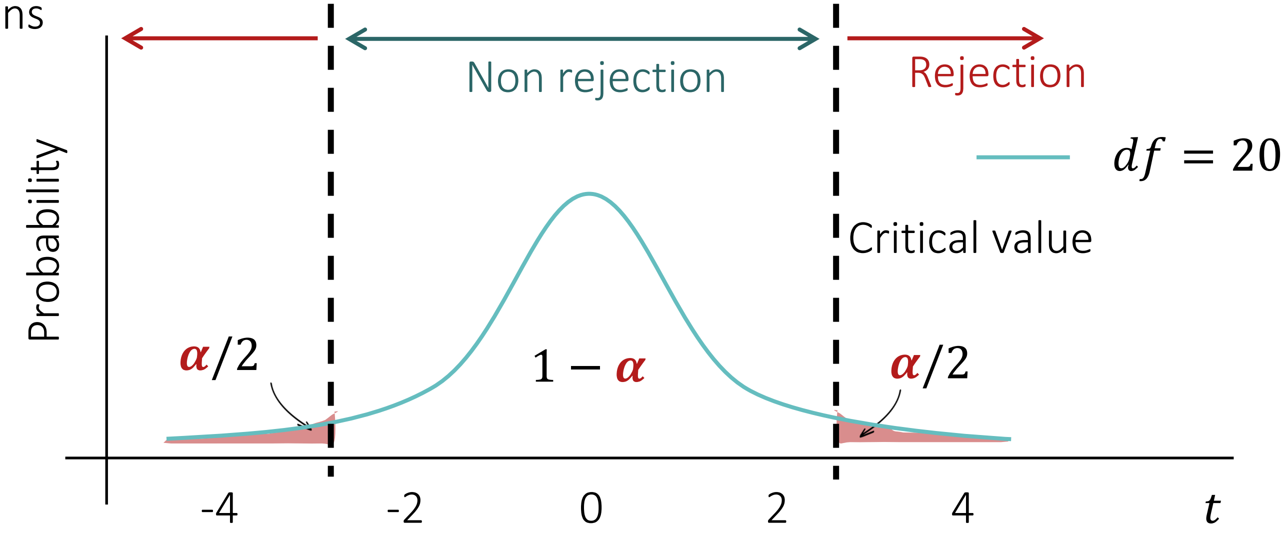
cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690

$t \to z$ when $df \to \infty$

$t \approx z$ when $df \geq 1{,}000$

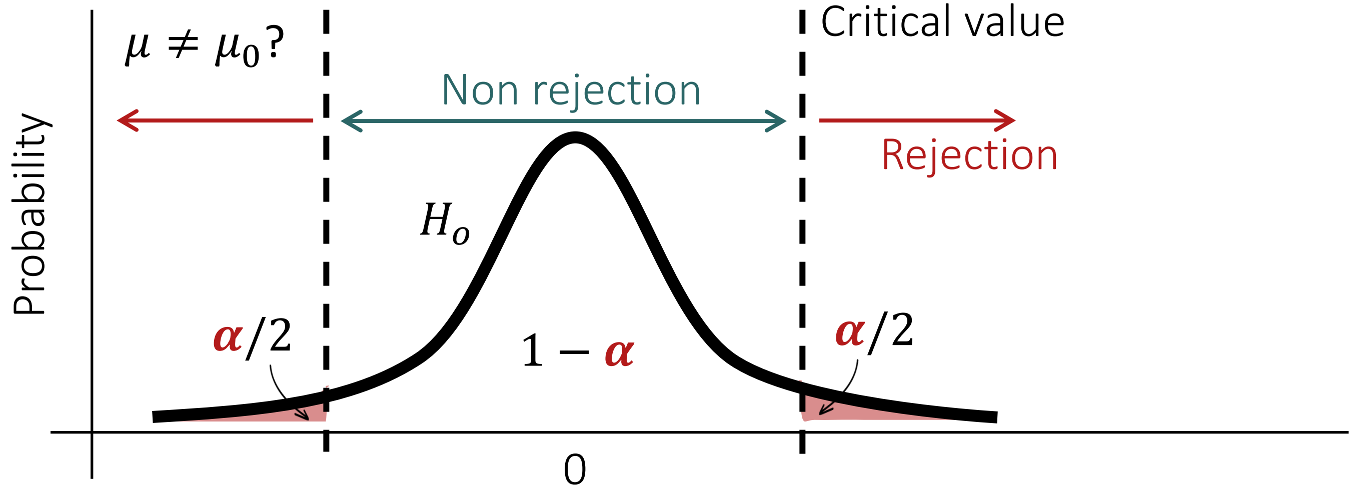
Known distributions

t-table

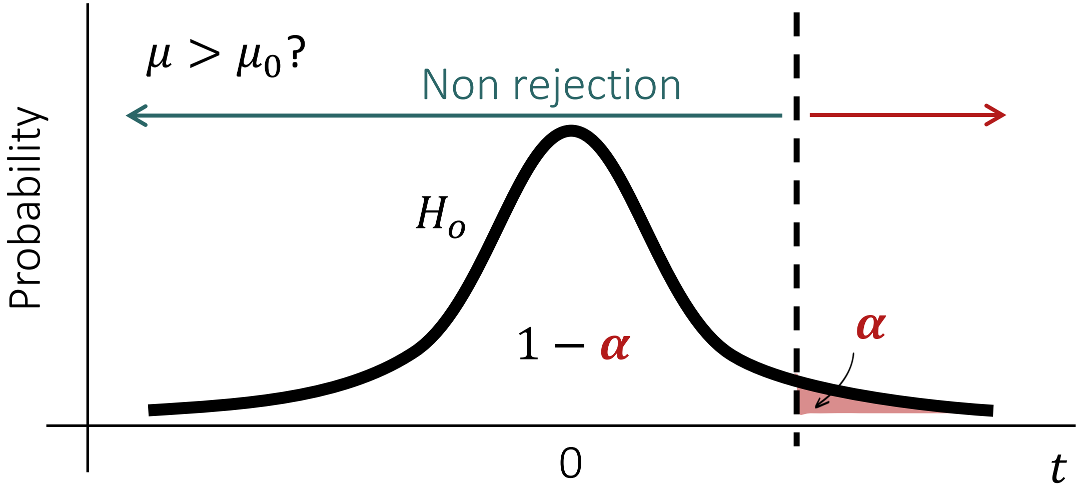


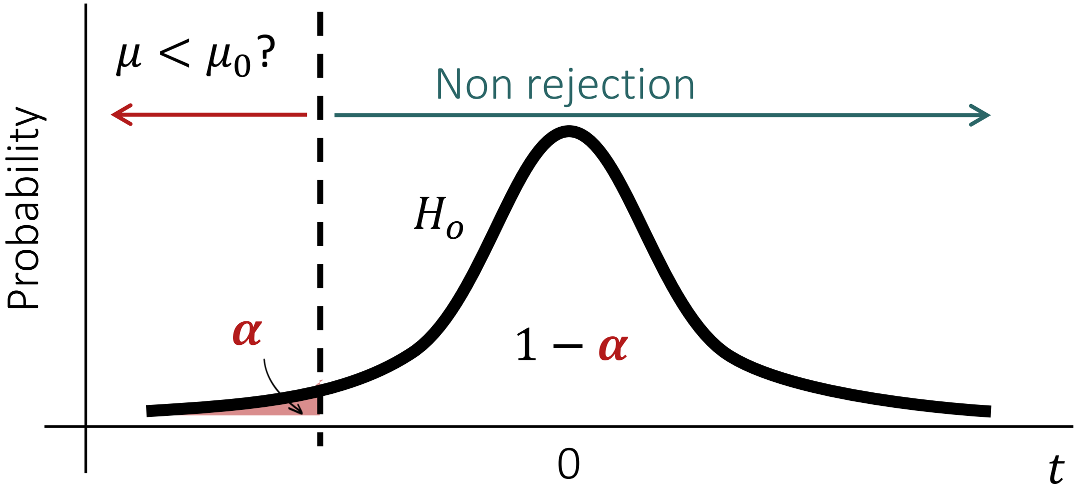
Statistical Tests: the T-test, How?

- Two-sided comparison ($\mu \neq \mu_0$?)



- One-sided comparison ($\mu > \mu_0$ or $\mu < \mu_0$?)





Statistical Tests: the T-test in Practice

We reject H_0

Japan and Ireland average daily calorie intakes are statistically different, with a level of confidence of 95%

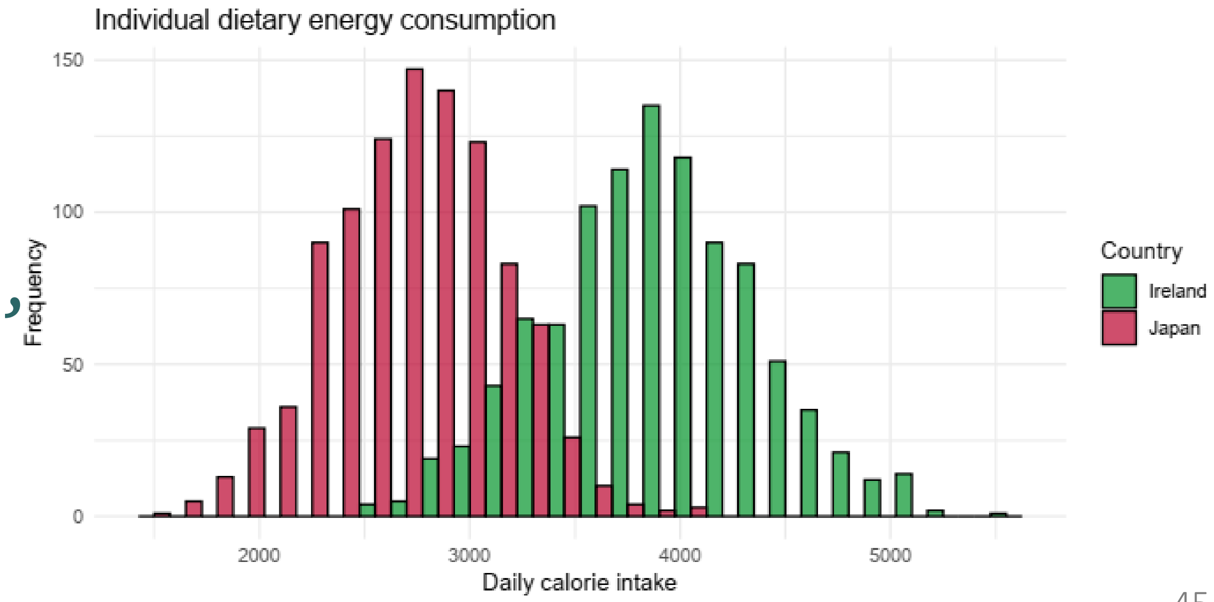
- Two-sample: 2 means together (μ_1 and μ_2)

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2 \frac{1}{n_1} - \frac{1}{n_2}}}, df = n_1 + n_2 - 2$$

With $\mu_1 - \mu_2 = 0$ under H_0, and $\bar{x}_1 - \bar{x}_2$ close to 0, hence z close to 0 if H_0 is true

- Is there a difference between Japan and Ireland average daily calorie intake?
- In R

```
t.test(data[data$country=="Japan",
      "calorie_intake"],
      data[data$country=="Ireland",
      "calorie_intake"])
t = -57.908, df = 1919.4,
p-value < 2.2e-16
```



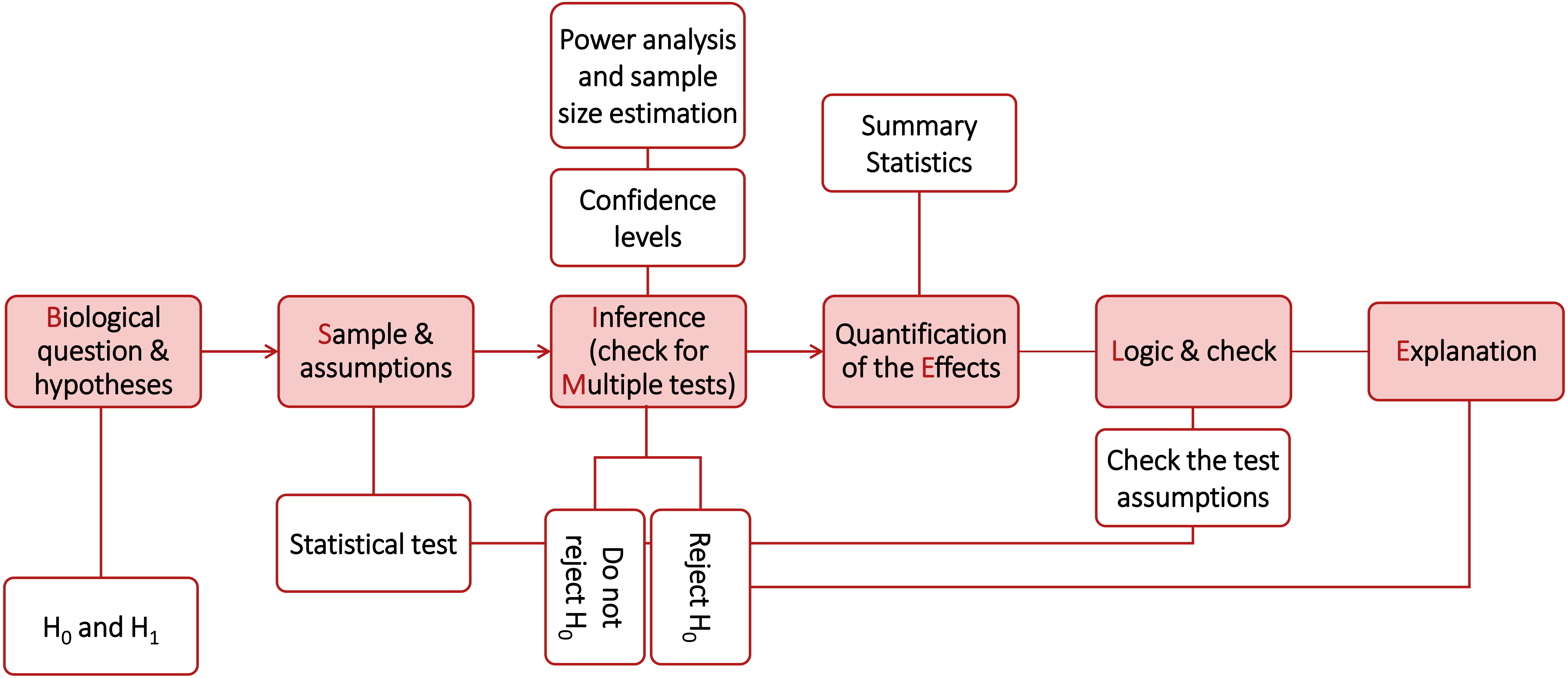
Statistical Tests: the T-test in Practice

- In R

```
t.test(x, y = NULL,
       alternative = c("two.sided", "less", "greater")
       mu = 0, paired = FALSE, var.equal = FALSE,
       conf.level = 0.95, ...)
```

Practice Time

Hypothesis Testing Workflow



Power and Sample Size

- Null hypothesis (H_0): absence of association between variables, i.e., absence of difference between groups ($\mu_1 = \mu_2$)
- Alternative hypothesis (H_1): association between variables, i.e., difference between groups ($\mu_1 \neq \mu_2$)

Null hypothesis (H_0) is…	True ($\mu_1 = \mu_2$)	False ($\mu_1 \neq \mu_2$)
Reject	Type I Error False positive **Probability of Type I Error =** $\boldsymbol{\alpha}$ **= significance**	Correct decision True positive (association) **Probability of no Type II Error =** $\mathbf{1} - \boldsymbol{\beta}$ **= power**
Fail to reject	Correct decision True negative (no association) **Probability of no Type I Error =** $\mathbf{1} - \boldsymbol{\alpha}$ **= confidence**	Type II Error False negative **Probability of Type II Error =** $\boldsymbol{\beta}$

Power and sample size

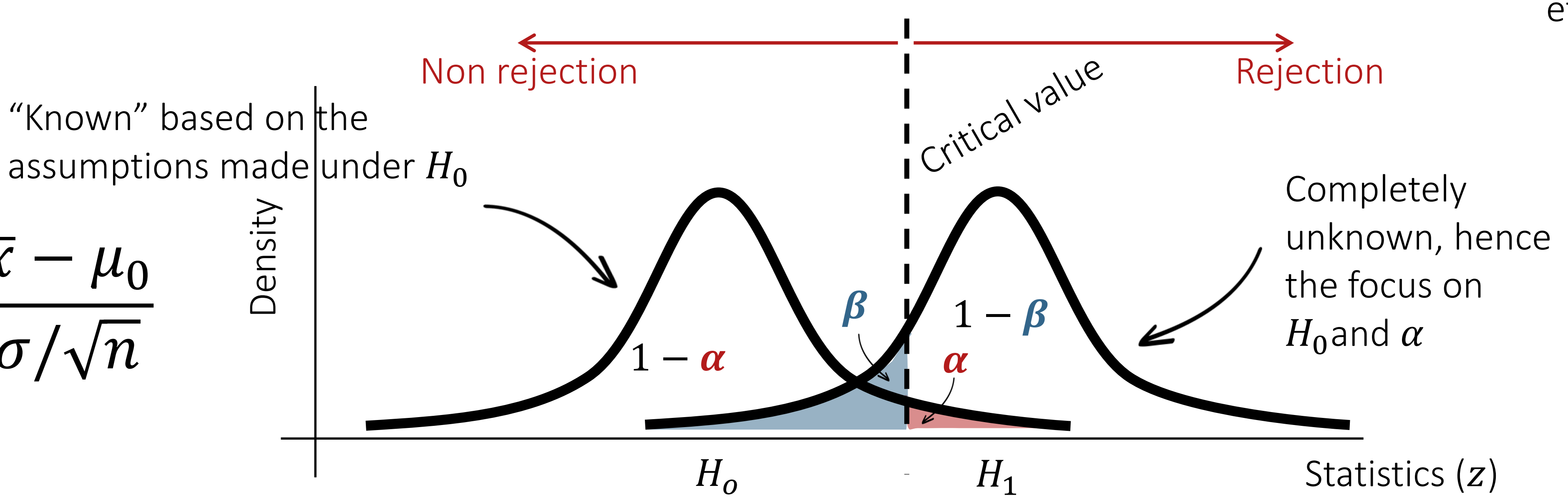
- Power: $1-\boldsymbol{\beta} = P(\text{reject } H_0 | H_0 \text{ is false}) = P(\text{no type II error})$
- Ability to detect associations when they exist

Needs assumptions on the true population means μ and how it differs from μ_0

- $1-\boldsymbol{\beta} = P\left(z > \frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{1-\alpha/2}\right) + P\left(z < \frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right)$
- We can find n giving the power $1-\boldsymbol{\beta}$ as $n = \left(\sigma \frac{z_{\alpha/2}+z_{\beta}}{\mu_0-\mu}\right)^2$

We can only calculate the power of analysis to detect **a specific difference** (called effect size)

$$z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$$



Power and Sample Size

- Power: $1 - \boldsymbol{\beta} = P(\text{reject } H_0 | H_0 \text{ is false}) = P(\text{no type II error})$
- Ability to detect associations when they exist
- $\nearrow n$ brings z further from $0 \rightarrow \nearrow$ the probability of detecting existing associations
- We cannot indefinitely $\nearrow n$ but we can identify the n needed to detect specified differences…

  Calculating the power of an analysis, or estimating the required sample size to detect a difference requires to know the difference we aim to detect (assumption, pilot data…)

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

z for the selected significance and power

$$n = \left(\sigma \frac{z_{\alpha/2} + z_{\beta}}{\mu_0 - \mu}\right)^2$$

Theoretical difference we aim to detect

What To Do When the Assumptions are Violated?

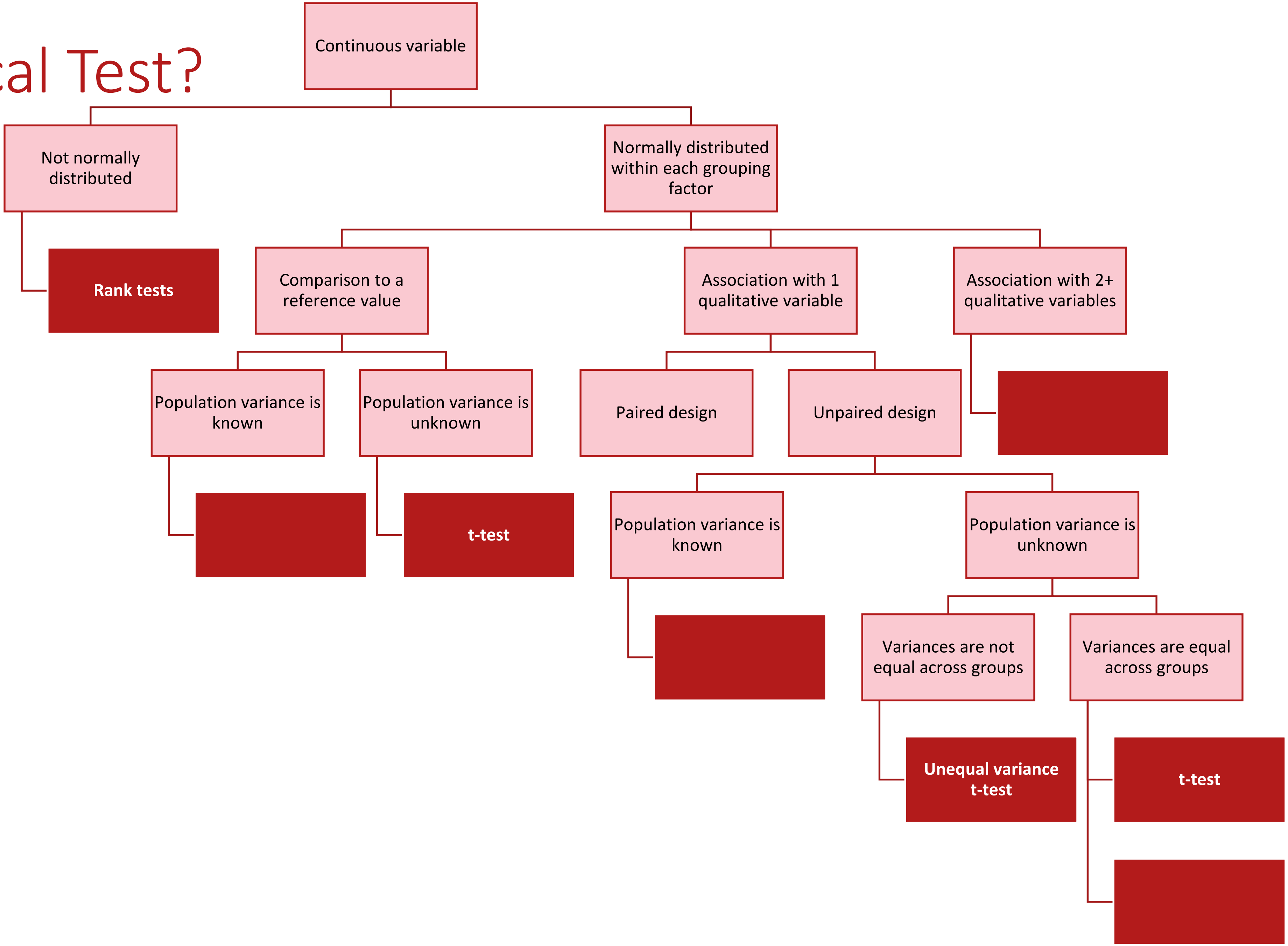
- Tests like the t-test that only apply when the data match a specific distribution are called **parametric tests**
- When this assumption is violated, we can use **non-parametric tests**

- Why do we bother with parametric tests? They have more statistical power (they are better at detecting associations that exist)

- **Rank tests** are common. Instead of looking at summary statistics, they rank observations to assess if one group tends to always be at the top, and one at the bottom

- Process is otherwise relatively similar

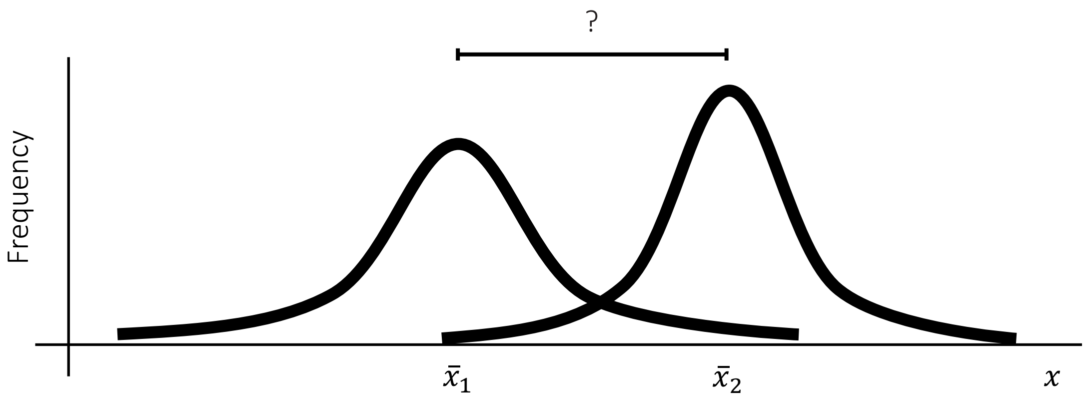
Which Statistical Test?



Practice Time

Comparison of Means

Paired-designs



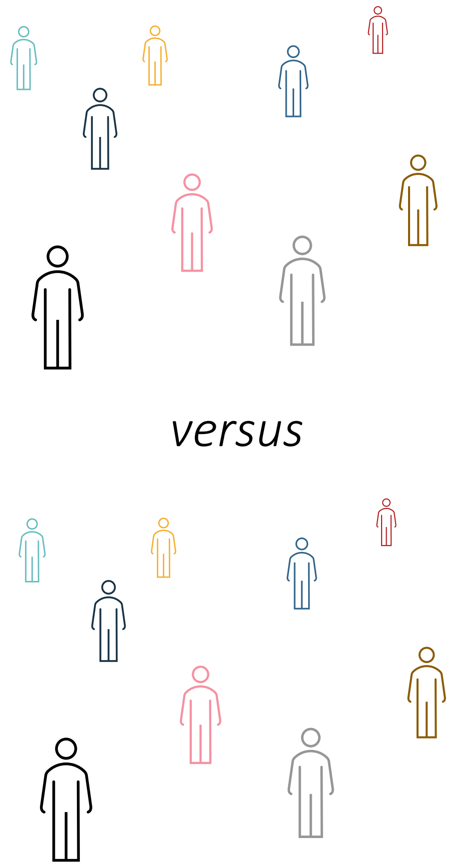
What Do We Mean by "Paired"?

Happens at the study-design stage, before data collection

- Pre-/post-intervention

Same group

Pre-intervention



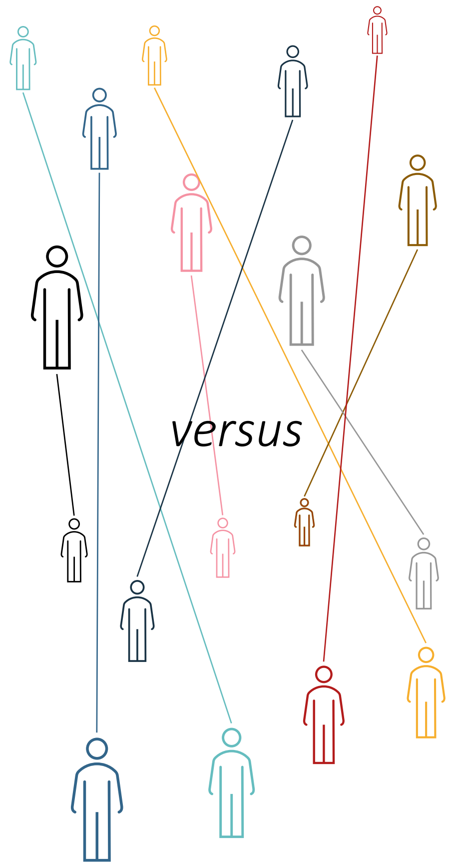
Post-intervention

Treated

↓

- Matched

Group A



Group B

Paired with

↓

Same individual characteristics (e.g., age, gender, ethnicity, socio-economic background…)

When e.g.:

- Cannot ethically voluntarily expose people (e.g., illicit drugs)
- Experiment is long

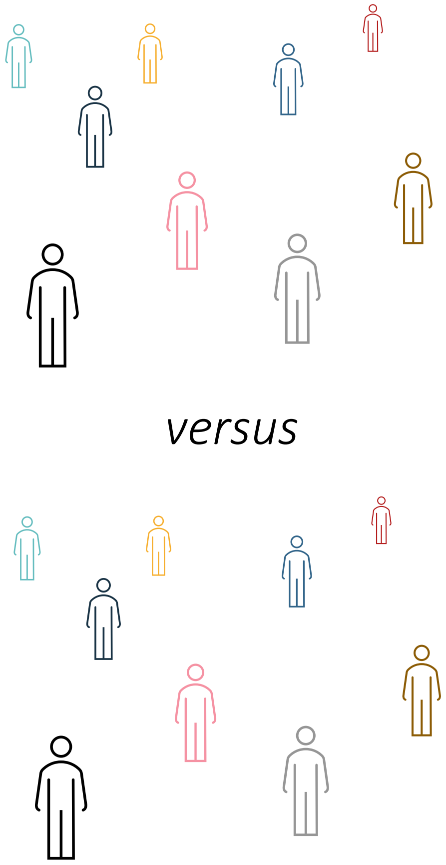
What Do We Mean by "Paired"?

Example – Assess the **safety and effectiveness** of an at-home, game-based **digital therapy** for treating adult patients with Attention-Deficit/Hyperactivity Disorder (**ADHD**)

- Pre-/post-intervention

Same group

Pre-intervention



Before starting the therapy

versus

Treated

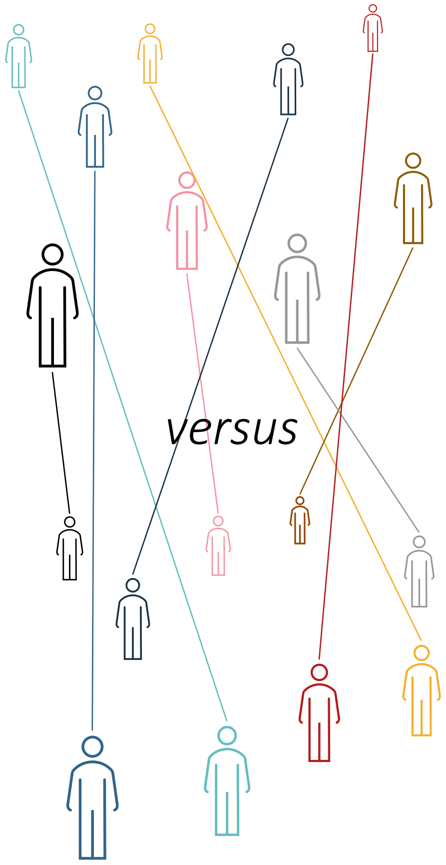
↓

Post-intervention

After X months of therapy

- Matched

Group A



Not exposed to the therapy

versus

Paired with

↓

Same individual characteristics (e.g., age, gender, ethnicity, socio-economic background…)

Group B

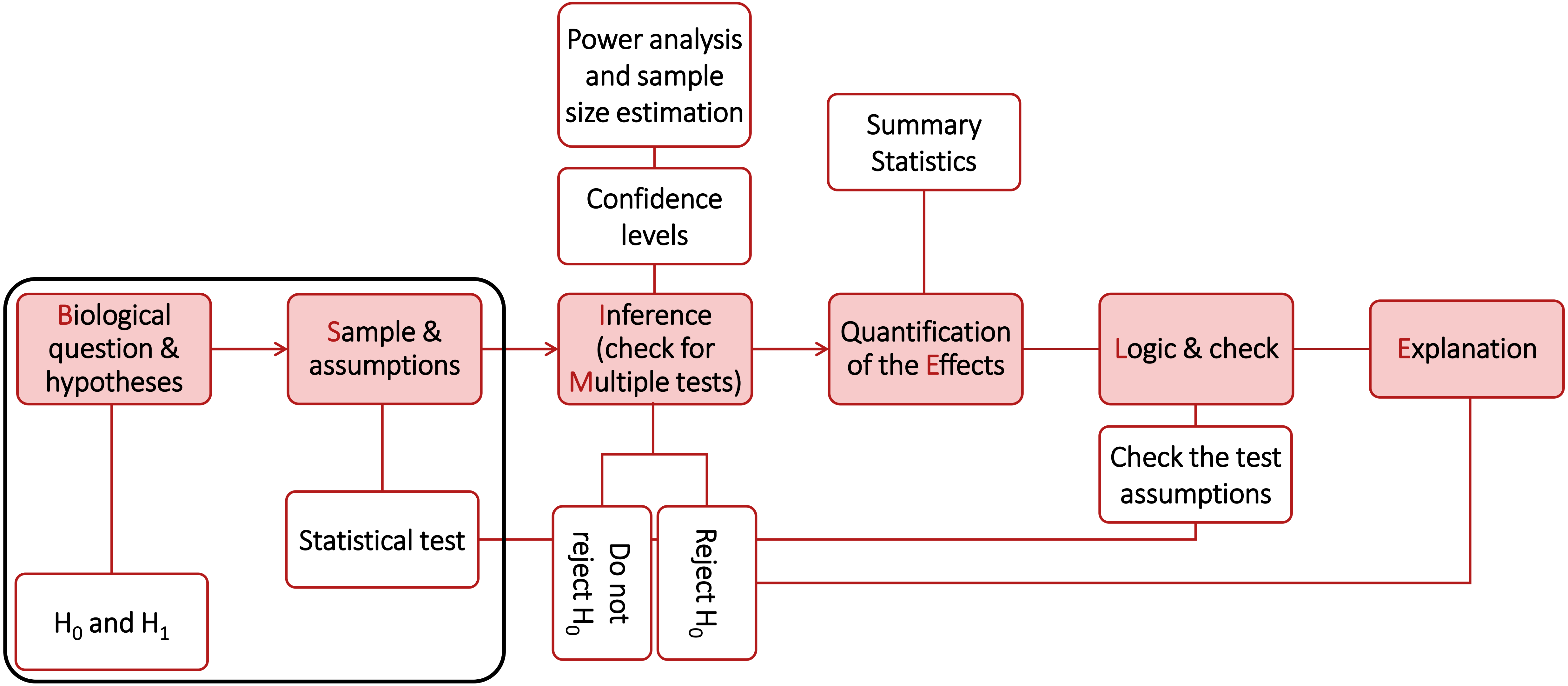
Exposed to the therapy

When e.g.:

- Cannot ethically voluntarily expose people (e.g., illicit drugs)
- Experiment is long

Inspired from an on-going clinical trial at University of California San Francisco

Hypothesis Testing Workflow



Working Hypotheses

- Null hypothesis (H_0): $\mu_1 = \mu_2$
  - ADHD symptom intensity is not associated with the treatment
  - Pre-/post-intervention design: there is no difference in pre- and post-treatment ADHD symptom intensity
  - Matched design: there is no difference in ADHD symptom intensity between the control and treated groups

- Alternative hypothesis (H_1): $\mu_1 \neq \mu_2$
  - ADHD symptom intensity is associated with the treatment
  - Pre-/post-intervention design: pre- and post-treatment ADHD symptom intensity are different
  - Matched design: ADHD symptom intensity is different between the control and treated groups

Study Design

Assess the **safety and effectiveness** of a **digital therapy** for treating adult patients with **ADHD**

- Data
  - Quantitative measure of AHDH

    E.g. "Quantified Behavior Test Plus"

    Inattention and impulsivity measured using a continuous performance test on a computer

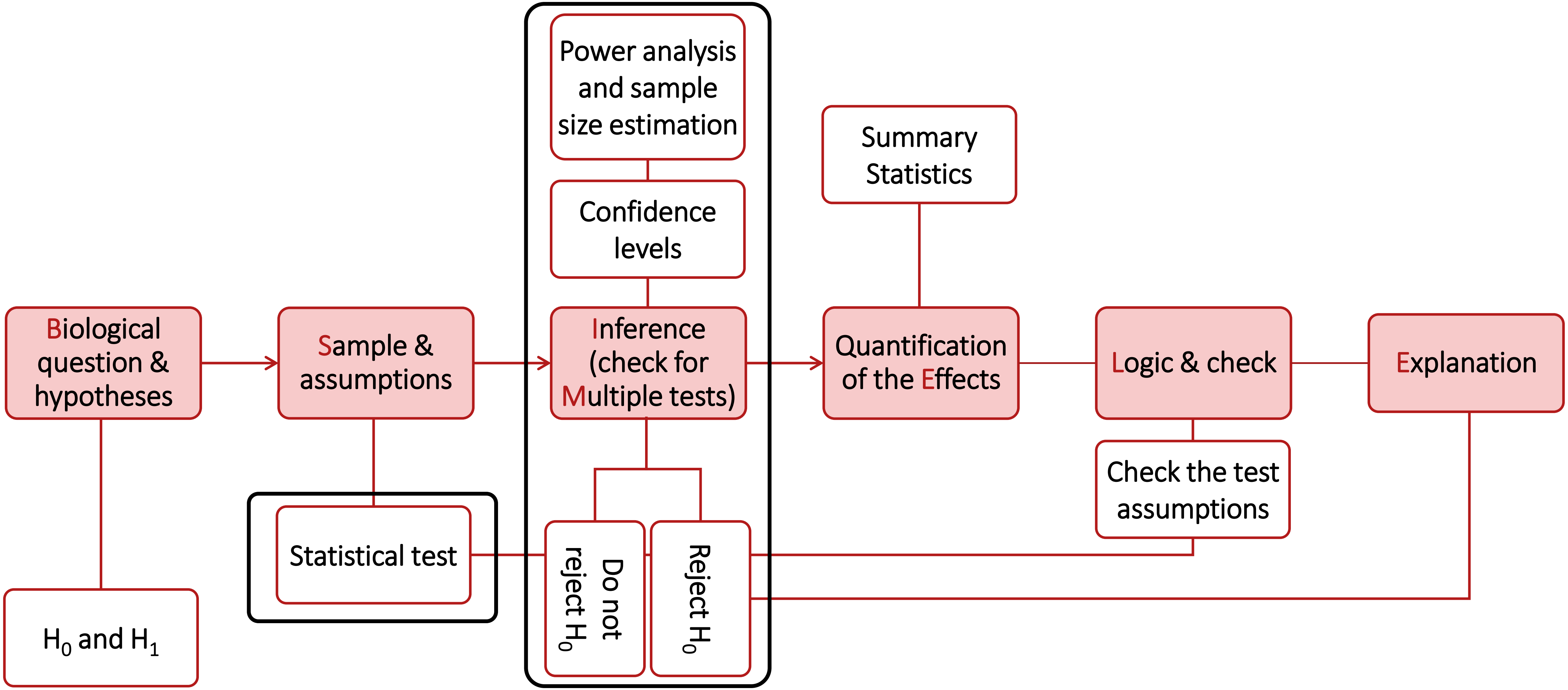
    Hyperactivity measured using a motion-tracking system

    → Continuous score
- Pairing procedure
  - Pre-/post-intervention

    This one here based on the protocol of the clinical trial (only ADHD participants)
  - Matched

Edebol et al, 2013. Measuring adult ADHD using the Quantified Behavior Test Plus. *PsyCh journal*.

Hypothesis Testing Workflow



Statistical Tests: the Paired T-test, Jow?

$* \sum(x_A - x_B)$

$** \sqrt{\frac{\sum((x_A - x_B) - \bar{x}_\Delta)^2}{n}}$

- Paired t-test

Observed mean difference within pairs

Expected difference under H_0 (usually 0)

Compared to a known distribution

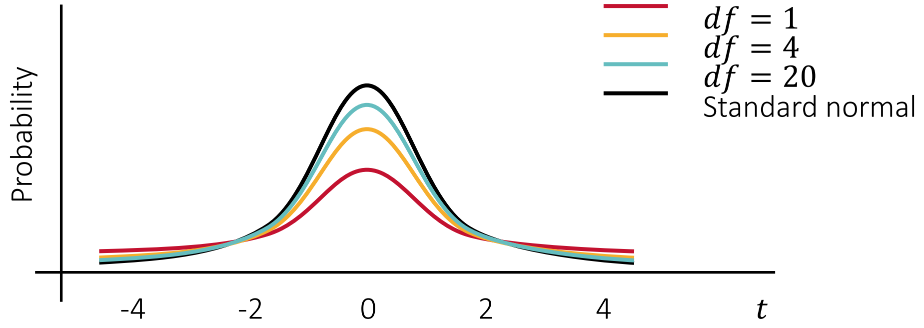
$$t = \frac{\bar{x}_\Delta - \mu_\Delta}{s_\Delta / \sqrt{n}}, df = n - 1$$

Standardized by variation

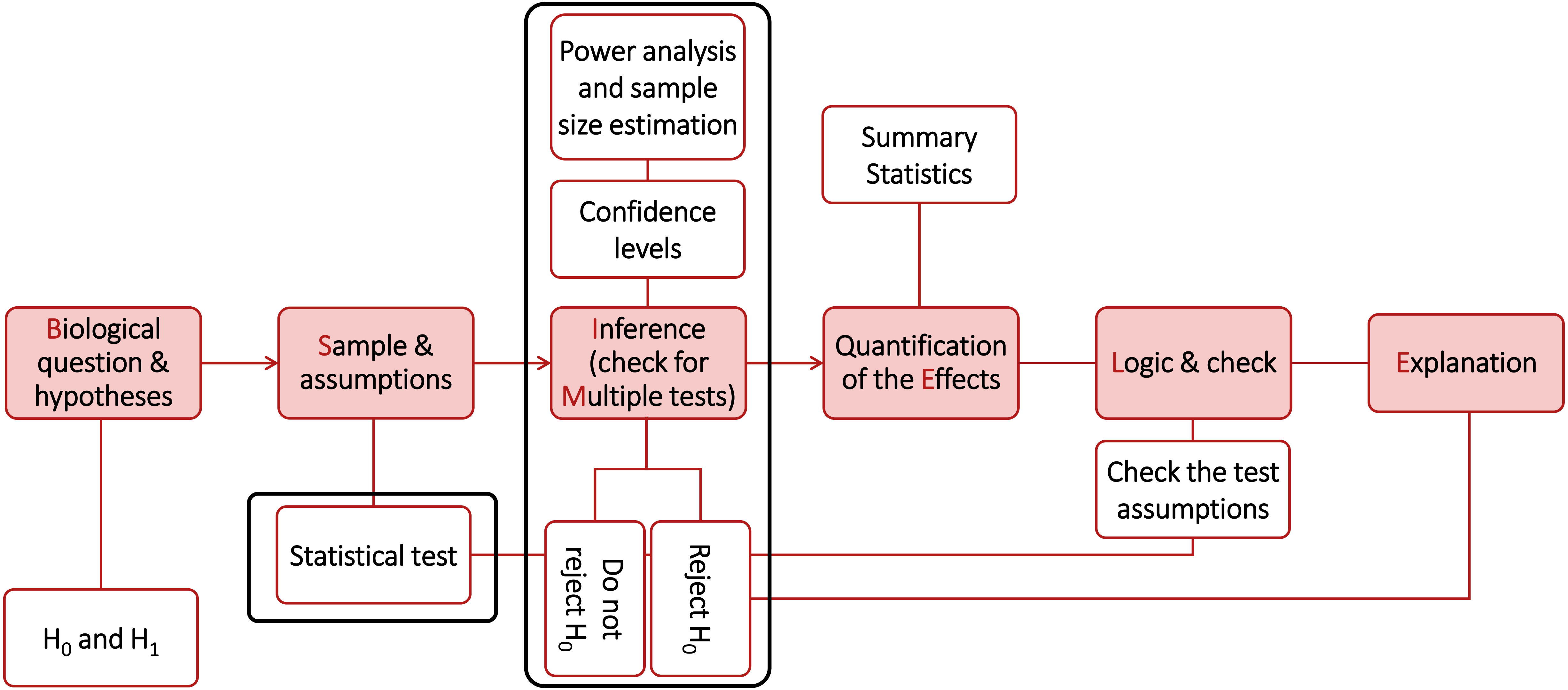
n is the sample size in each group (and the same in both groups), $\bar{x}_\Delta$ is the mean difference*, s_Δ is the standard deviation of the difference**

= One-sample t-test on the mean difference

With $\bar{x}_\Delta$ close to μ_Δ, hence t close to 0 if H_0 is true

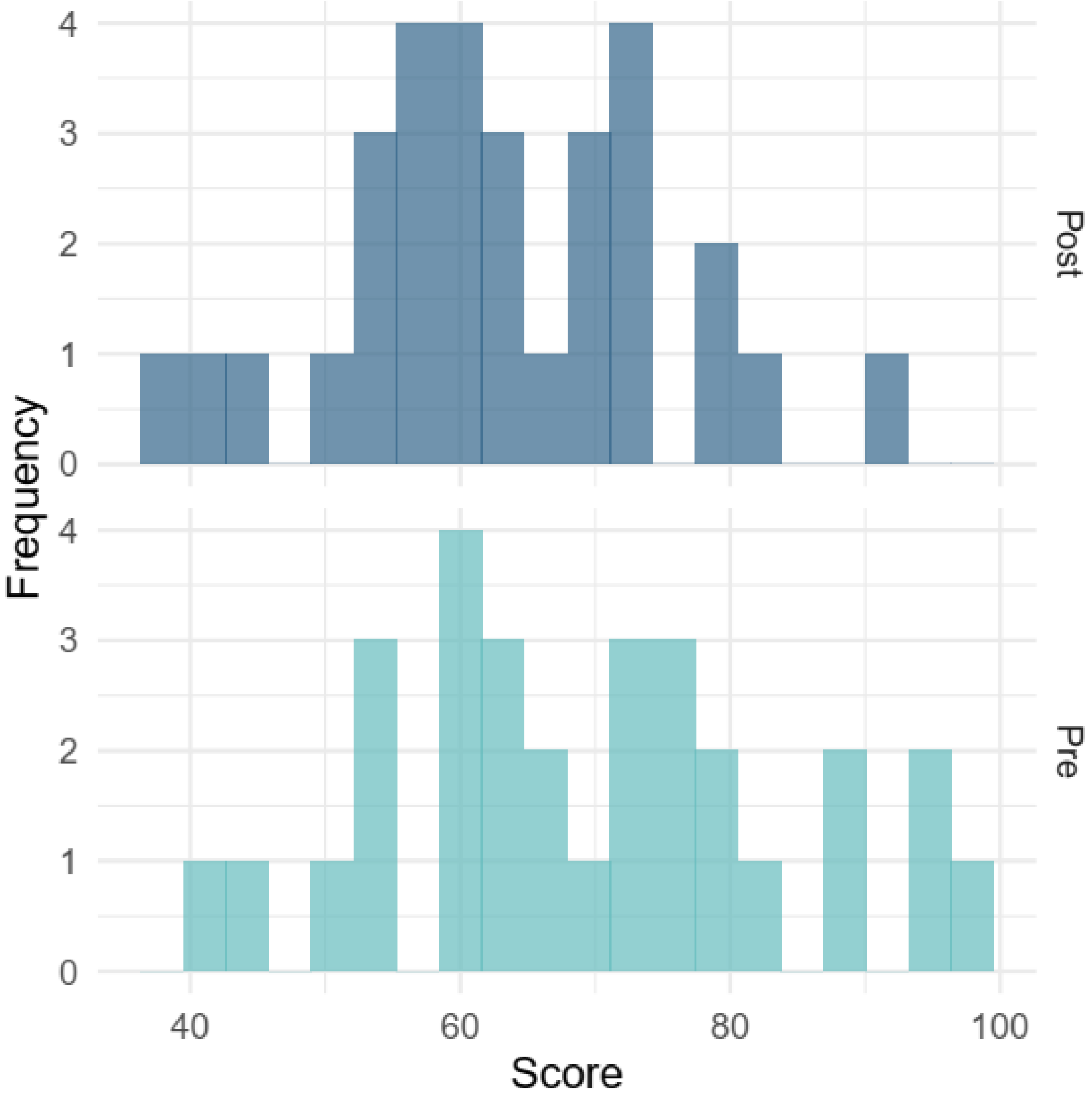


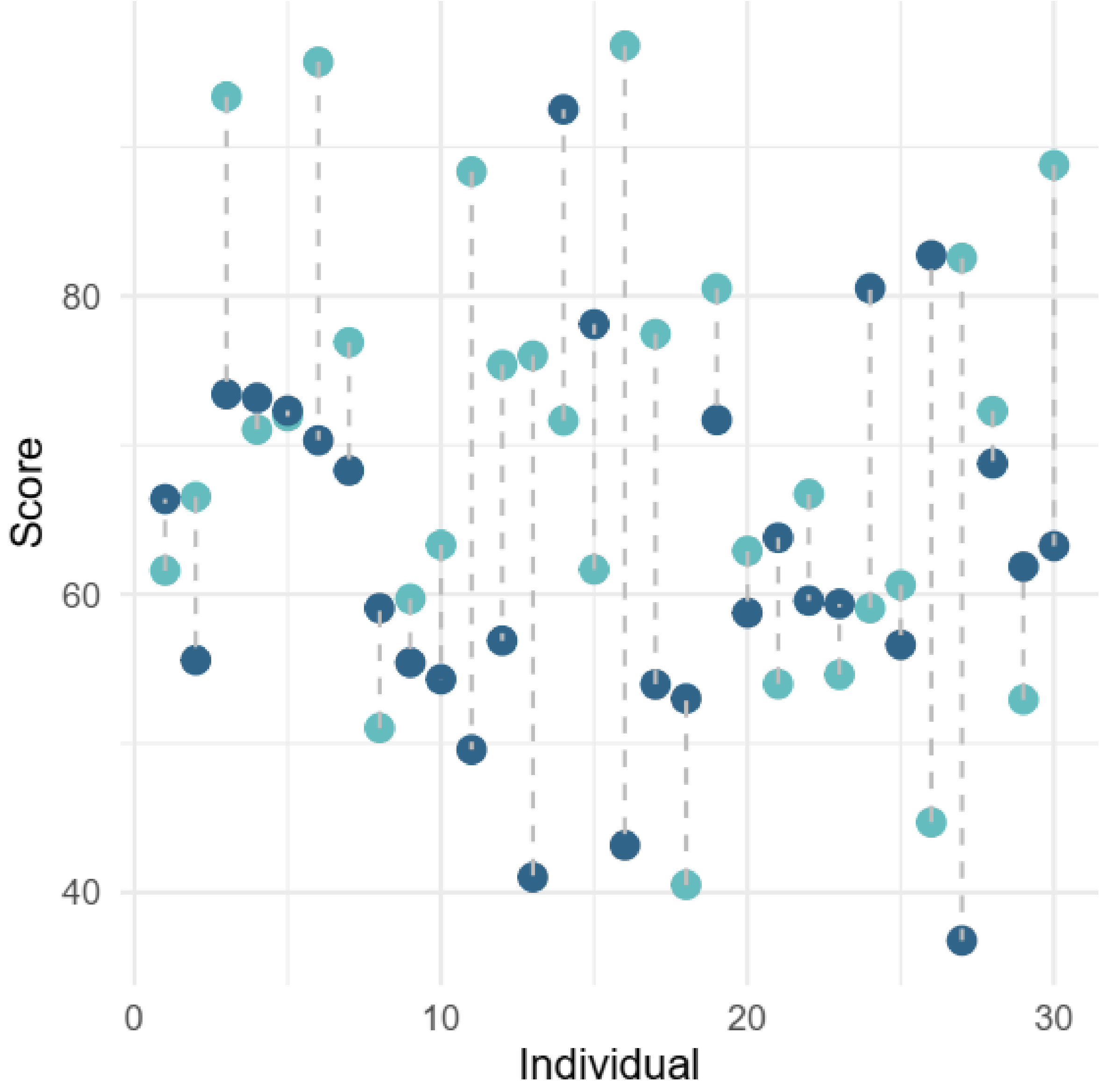
Hypothesis Testing Workflow



First Exploration of the Data

Assess the **safety and effectiveness** of a **digital therapy** for treating adult patients with **ADHD**

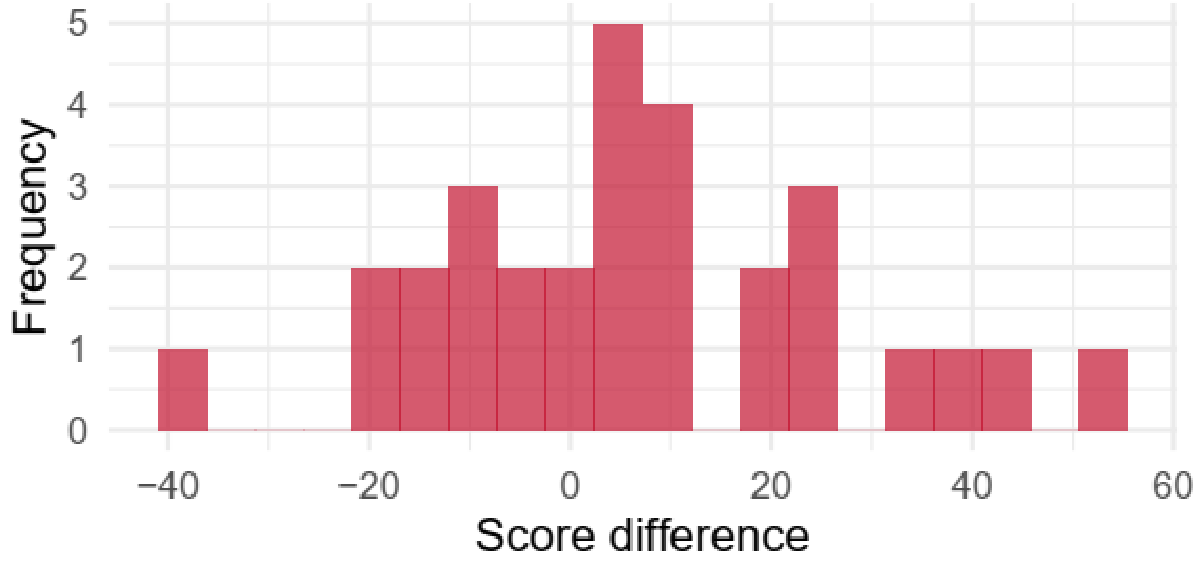


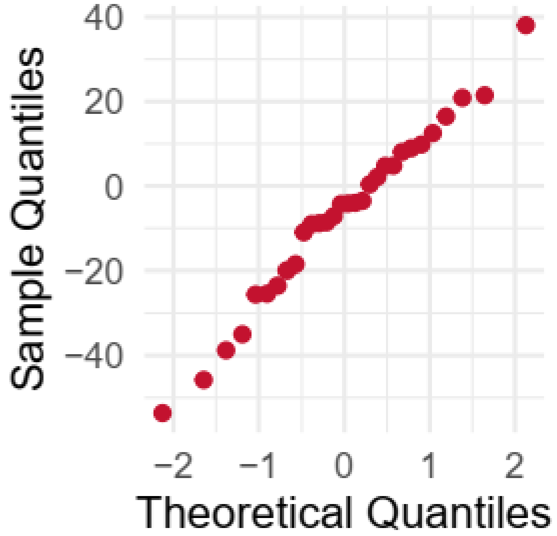


Statistical Tests: the Paired T-test, When?

Assess the **safety and effectiveness** of a **digital therapy** for treating adult patients with **ADHD**

- Assumptions on the data
  - **Paired measurements** are obtained from the same individual or from matched individuals — Study design
  - Pairs are **independent** from one another — Study design
  - <u>Difference</u> are **normally distributed** — Visual checks or statistical tests






```
Shapiro-Wilk
normality test data:
W = 0.98267,
p-value = 0.8913
```

Practice Time

Statistical Tests: the Paired T-test in Practice

Easily doable by hand

- Paired t-test

Observed mean difference within pairs

Expected difference under H_0 (usually 0)

Compared to a known distribution

$$t = \frac{\bar{x}_\Delta - \mu_\Delta}{s_\Delta / \sqrt{n}}, df = n - 1$$

Standardized by variation

n is the sample size in each group (and the same in both groups), $\bar{x}_\Delta$ is the mean difference*, s_Δ is the standard deviation of the difference**

= One-sample t-test on the mean difference

With $\bar{x}_\Delta$ close to μ_Δ, hence t close to 0 if H_0 is true $\rightarrow$ Compare the t-table values

Statistical Tests: the Paired T-test in Practice

Even easier in R

- Paired t-test

Observed mean difference within pairs

Expected difference under H_0 (usually 0)

Compared to a known distribution

$$t = \frac{\bar{x}_\Delta - \mu_\Delta}{s_\Delta / \sqrt{n}}, df = n - 1$$

Standardized by variation

n is the sample size in each group (and the same in both groups), $\bar{x}_\Delta$ is the mean difference*, s_Δ is the standard deviation of the difference**

= One-sample t-test on the mean difference

```
t.test(x, y, paired = TRUE)
```

Statistical Tests: the Paired T-test in Practice

Power was ≈ 90%... We got unlucky?

- Null hypothesis (H_0): $\mu_1 = \mu_2$
  - There is no difference in pre- and post-treatment ADHD symptom intensity

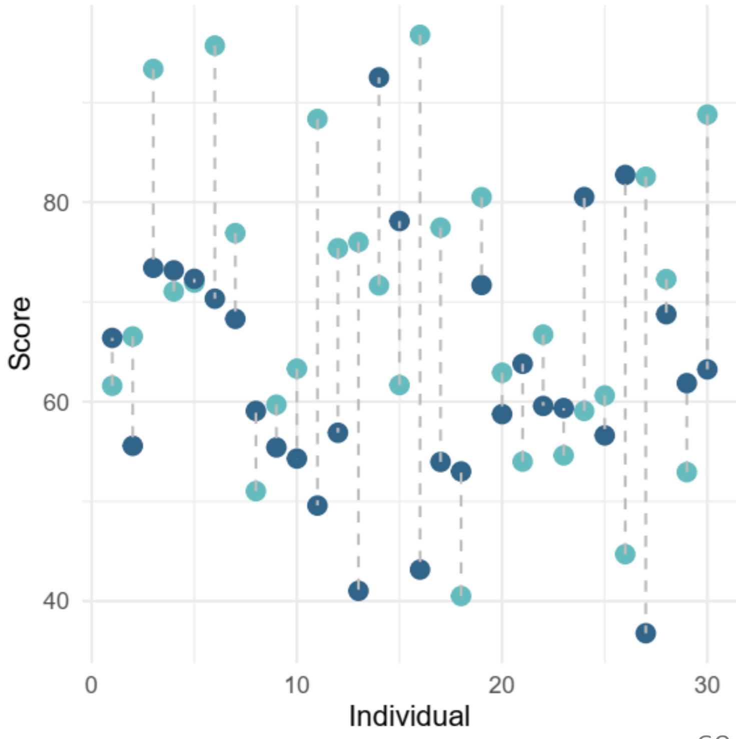
```
t.test(x = paired_data$Pre, y = paired_data$Post, paired = TRUE)
```

```
t = 1.7451, df = 29, p-value = 0.09156
alternative hypothesis: true mean difference
is not equal to 0
95 percent confidence interval:
-1.138395 14.375133
sample estimates:
mean difference
6.618369
```

$p > 0.05$

We do not reject H_0

We did not detect a difference in pre- and post-treatment ADHD symptom intensity



Assess the **safety and effectiveness** of a **digital therapy** for treating adult patients with **ADHD**

Statistical Tests: the Paired T-test in Practice

Changing the seed (new simulation, same parameter)

- Null hypothesis (H_0): $\mu_1 = \mu_2$
  - There is no difference in pre- and post-treatment ADHD symptom intensity

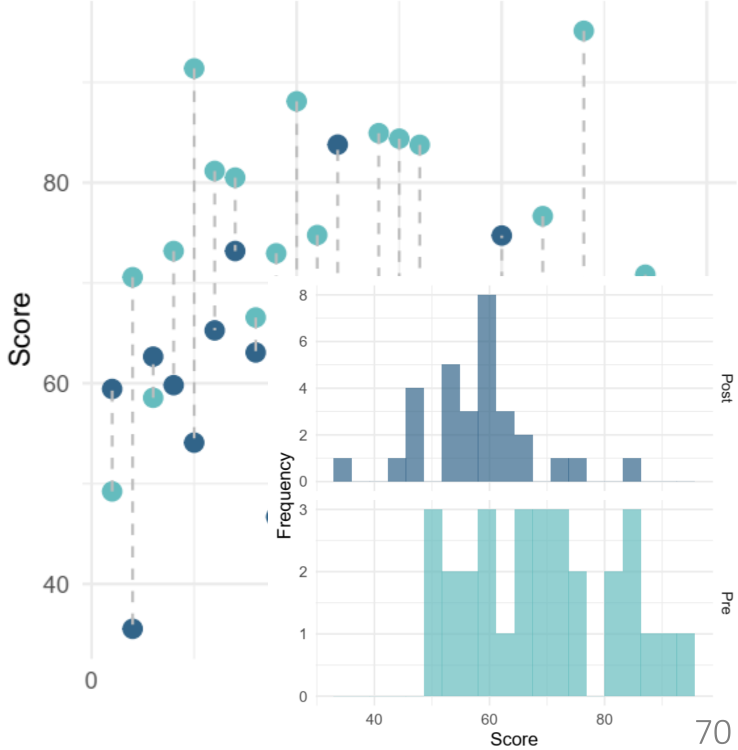
```
t.test(x = paired_data$Pre, y = paired_data$Post, paired = TRUE)
```

```
t = 3.7665, df = 29, p-value = 0.0007512
alternative hypothesis: true mean difference
is not equal to 0
95 percent confidence interval:
5.251354 17.730442
sample estimates:
mean difference
11.4909
```

$p < 0.05$

We reject H_0

There is a difference in pre- and post-treatment ADHD symptom intensity



Experiment → causal
There is an effect of treatment

Assess the **safety and effectiveness** of a **digital therapy** for treating adult patients with **ADHD**

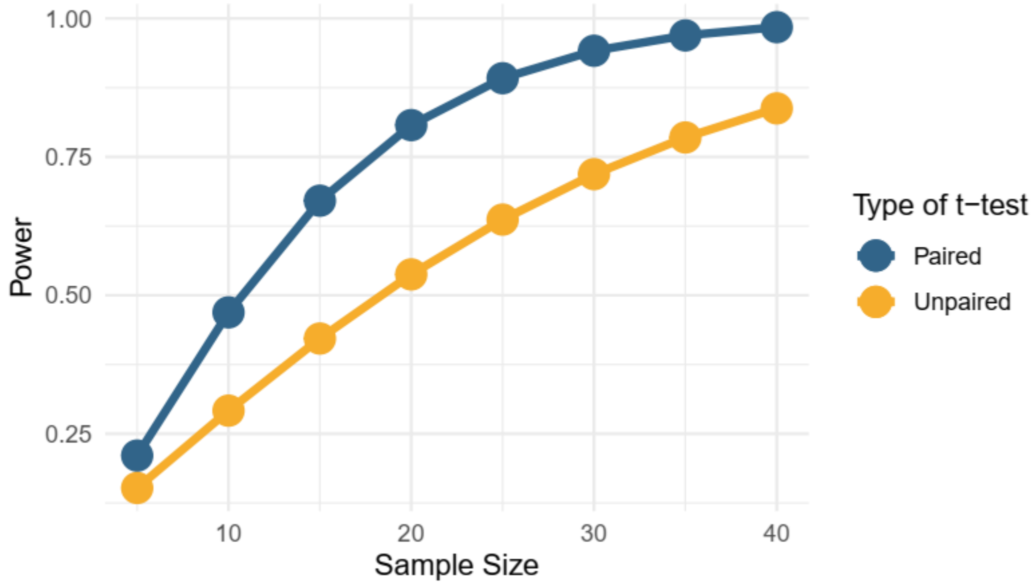
Paired Designed and Power

Assess the **safety and effectiveness** of a **digital therapy** for treating adult patients with **ADHD**

- Confidence & significance levels
  - "We used an α level of .05 for all statistical tests"

- Power analysis
  - **Expected difference** in ADHD symptom intensity score: 10
    (Based on the fact that difference in ADHD symptom intensity score between AHDH and non-ADHD patients was estimated at 20)
  - **Variance**: 15
    (Based on ADHD symptom intensity score variance in AHDH patients)
  - **Acceptable power**: 90%

```
power.t.test(n = nn, delta = 10, sd = 15,
sig.level = 0.05, type = "paired", alternative
= "two.sided")
```

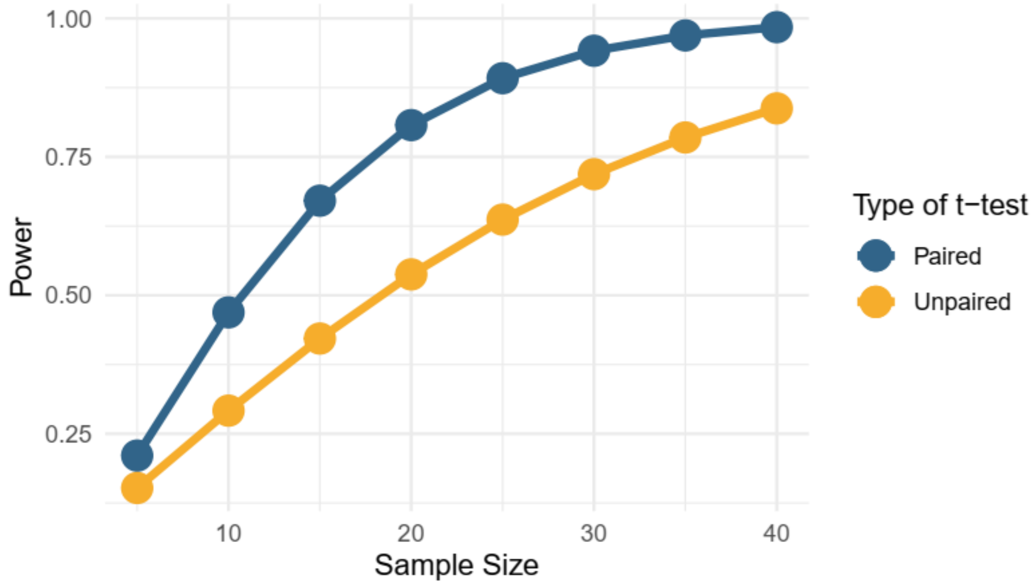


Paired Designed and Power

Assess the **safety and effectiveness** of a **digital therapy** for treating adult patients with **ADHD**

- **Paired tests are always more powerful** than their unpaired version
- Intuitive explanation(s)
  - A change in score in the same after treatment is more meaningful than an overall difference across two groups
  - There is less noise when focusing on paired samples than random samples

```
power.t.test(n = nn, delta = 10, sd = 15,
sig.level = 0.05, type = "paired", alternative
= "two.sided")
```



Paired Designed and Power

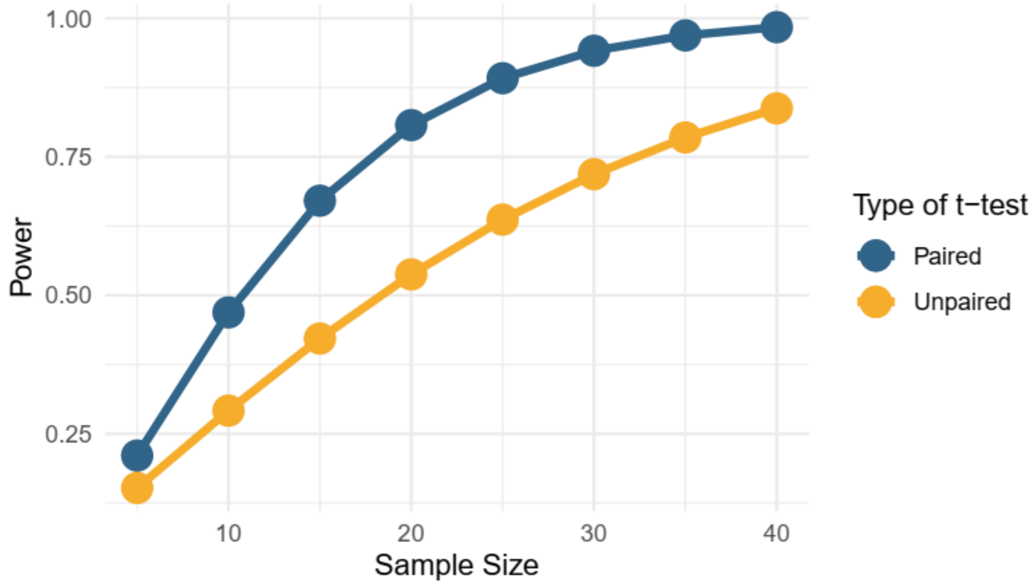
Assess the **safety and effectiveness** of a **digital therapy** for treating adult patients with **ADHD**

- Sample size estimation
  - **Expected difference** in ADHD symptom intensity score: 10
    (Based on the fact that difference in ADHD symptom intensity score between AHDH and non-ADHD patients was estimated at 20)
  - **Variance**: 15
    (Based on ADHD symptom intensity score variance in AHDH patients)
  - **Acceptable power**: 90%

Sample size required

- Paired t-test: 26
- Unpaired t-test: 48

```r
power.t.test(power = 0.90, delta = 10, sd = 15, sig.level = 0.05, type = "paired", alternative = "two.sided")
```

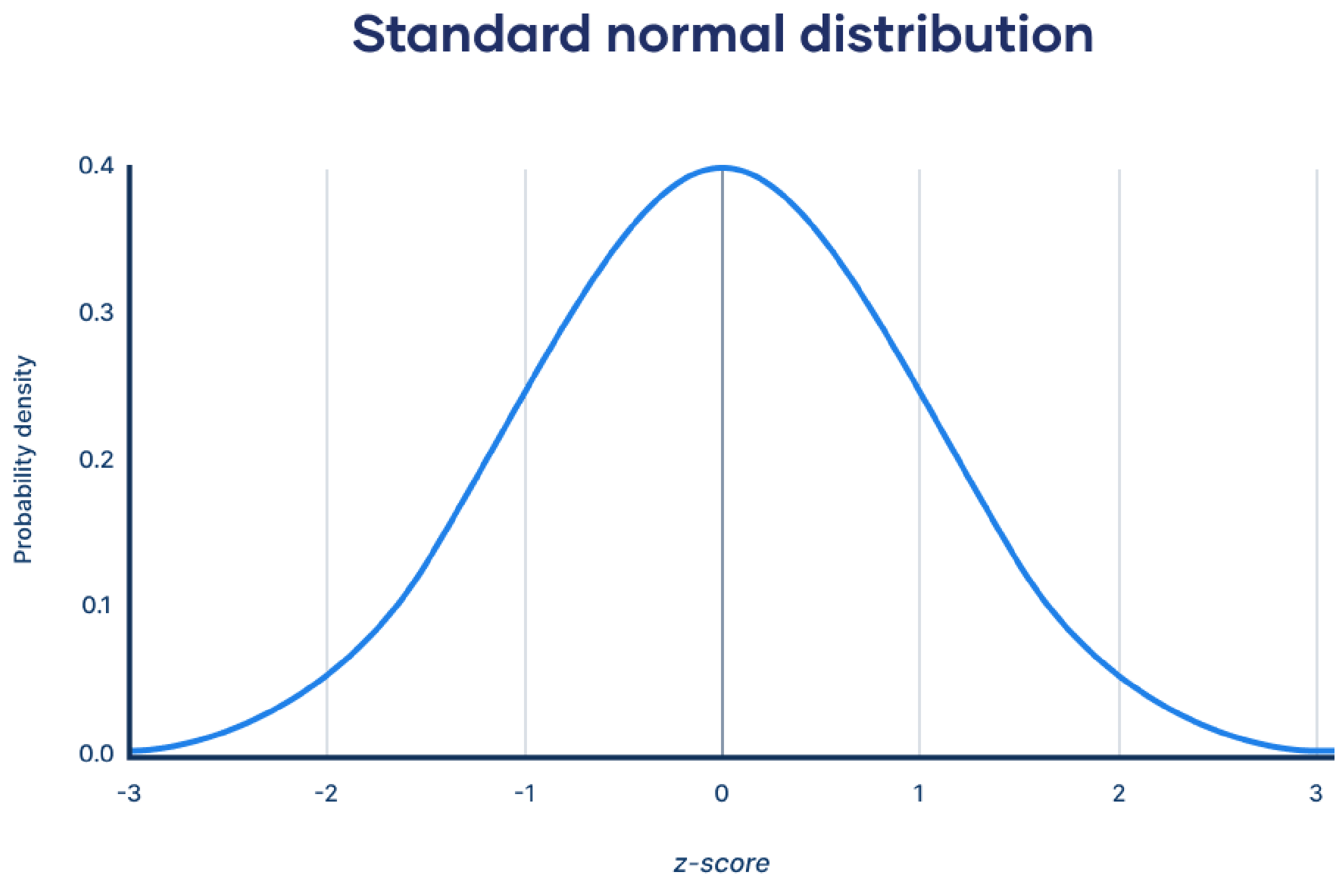


Behind the Scene: Where the Numbers Are Coming from…

The Z Distribution

Deviations from the Norm: Z Distribution and Z Score

- Z distribution = normal distribution with mean = 0 and standard deviation = 1



The Normal Distribution, the Z Distribution, and Intervals

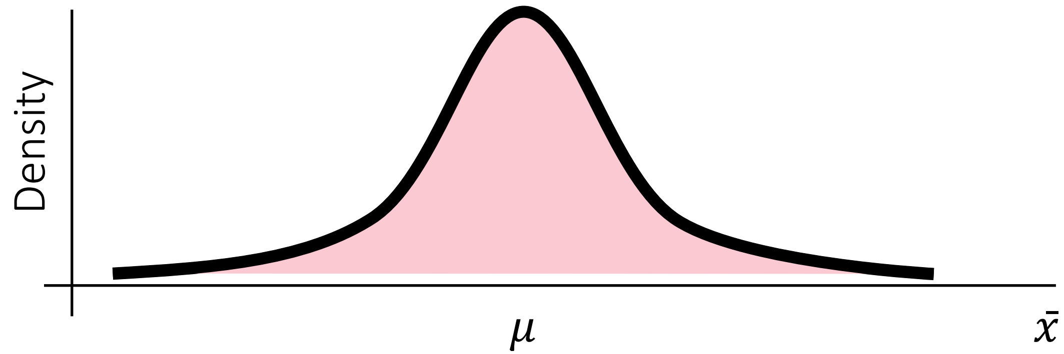
- Challenge: we do not know the position of the population mean $\boldsymbol{\mu}$
- Good news: we know the distribution of the sample mean $\bar{\boldsymbol{x}}$
- PDF and CDF of a normal distribution

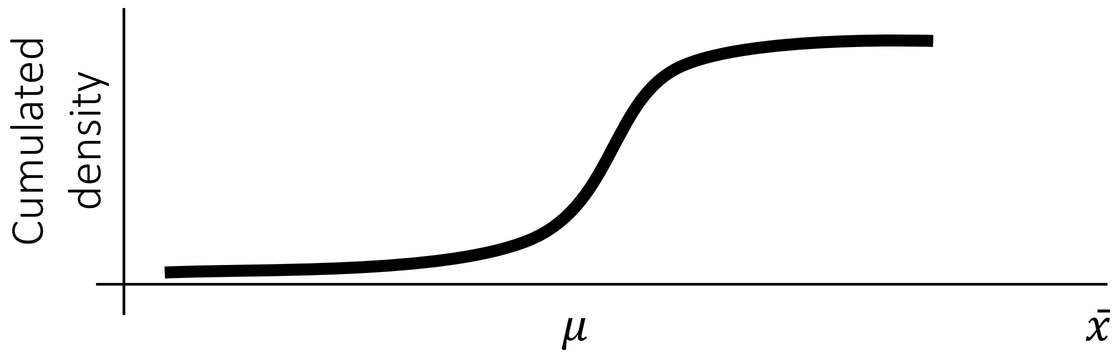
Integral for all the potential values of a continuous variable

$$\boldsymbol{N(\mu,\sigma)} \qquad P(\bar{x} = x) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad P(\bar{x} \leq x) = \frac{1}{\sigma\sqrt{\pi}} \int_{-\infty}^{x} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- PDF and CDF of the Z distribution

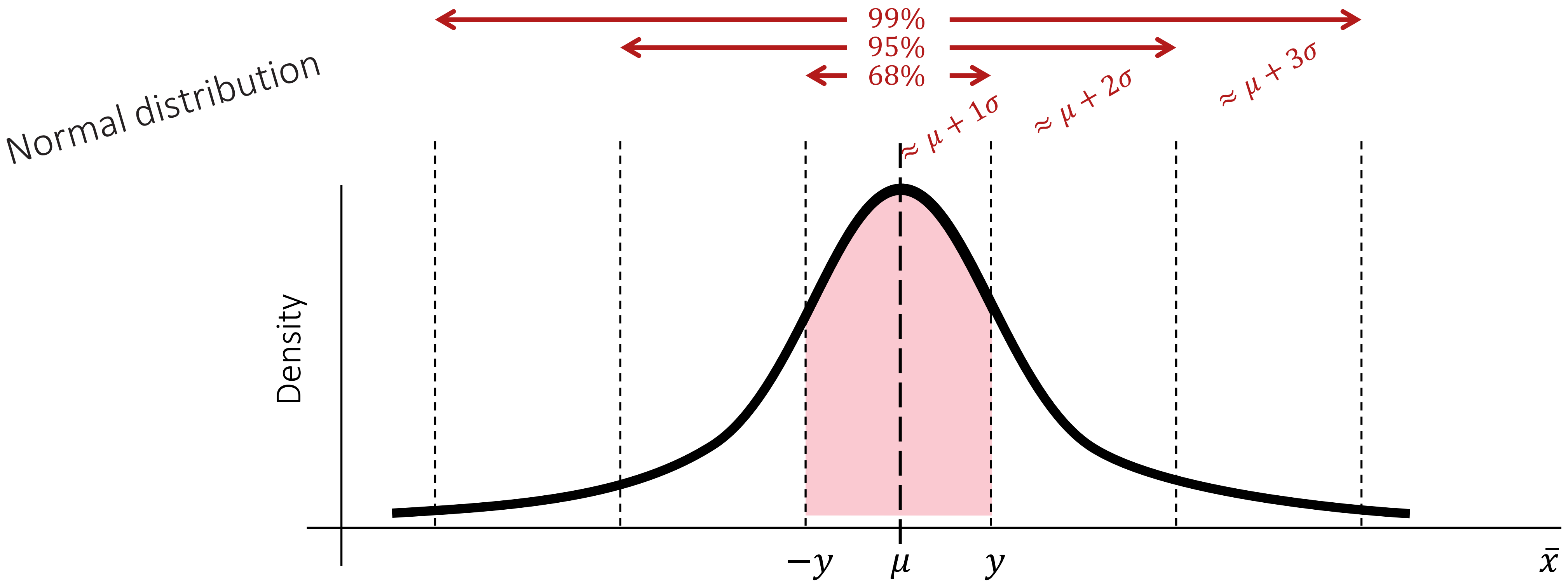
$$\boldsymbol{N(0,1)} \qquad P(\bar{z} = z) = \frac{1}{\sqrt{\pi}} e^{-\frac{z^2}{2}} \qquad P(\bar{z} \leq z) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{z} e^{-\frac{z^2}{2}} dz$$





The Normal Distribution, the Z Distribution, and Intervals

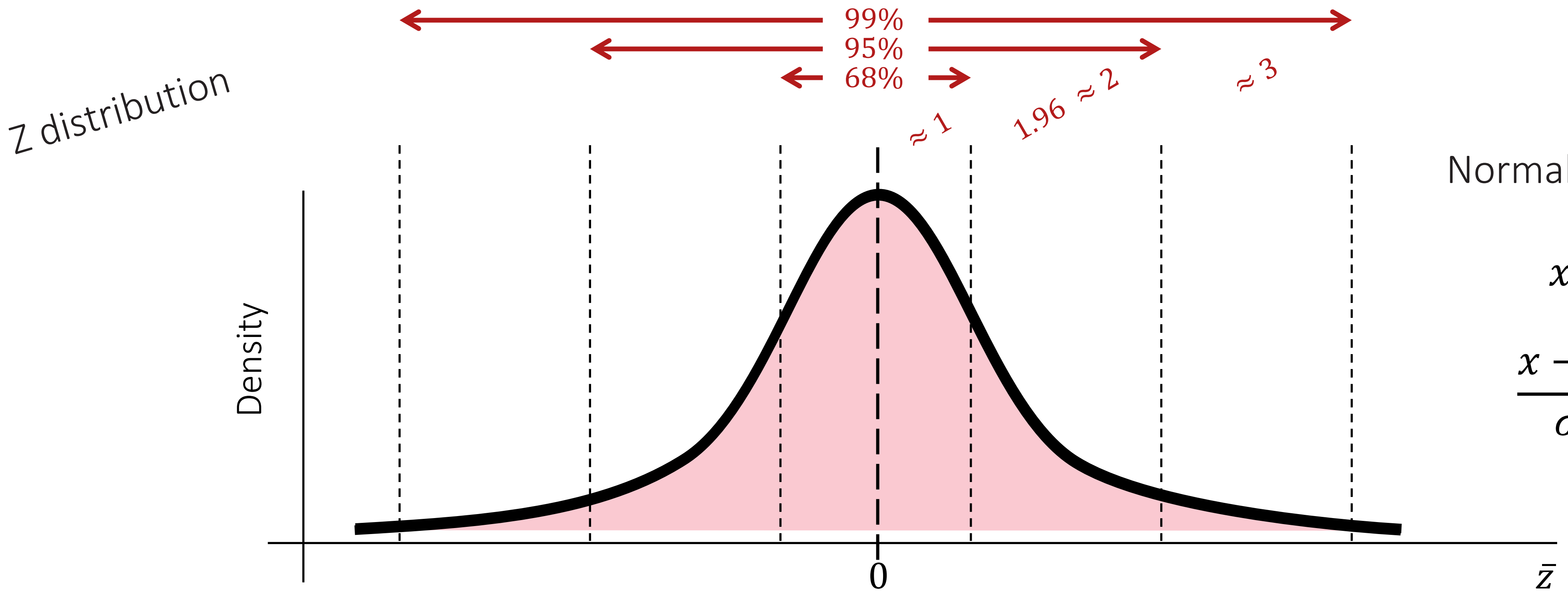
- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$



$$P(-y \leq \bar{x} \leq y) = \frac{1}{\sigma\sqrt{\pi}} \int_{-y}^{y} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The Normal Distribution, the Z Distribution, and Intervals

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$



Normal → Z distribution

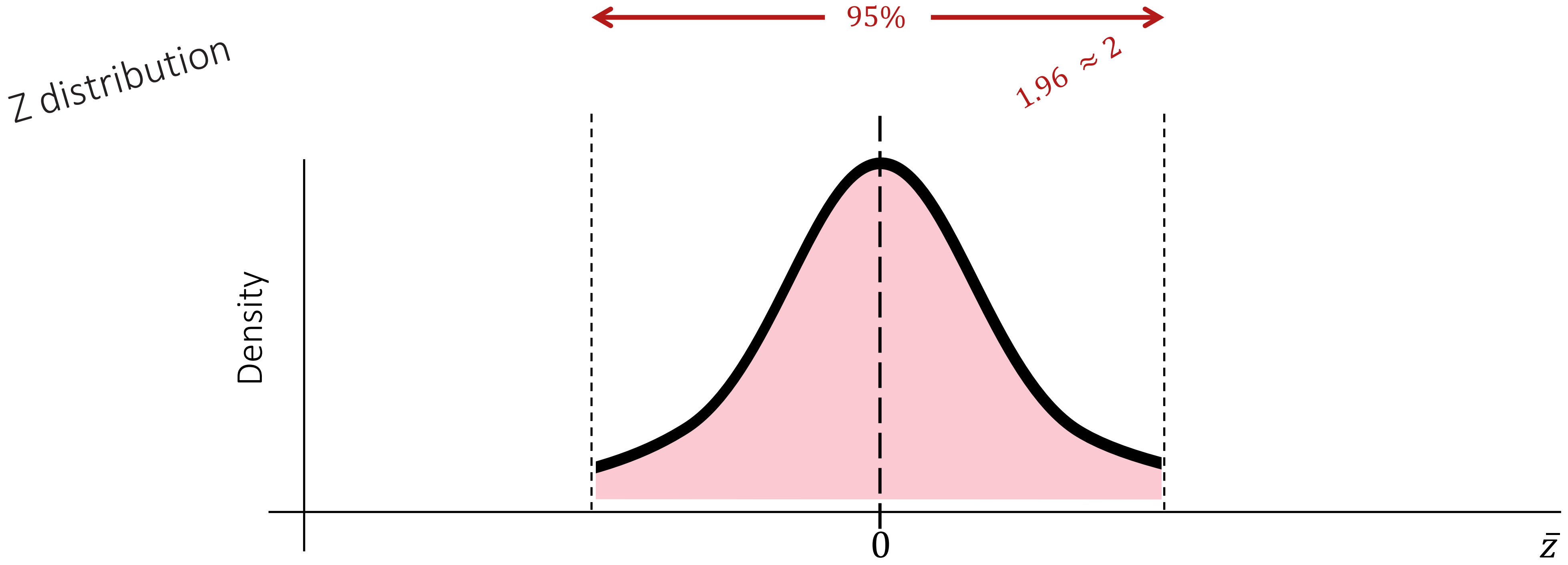
$$x \sim N(\mu, \sigma)$$

$$\frac{x-\mu}{\sigma} \sim N(0,1)$$

$$P(-y \leq \bar{z} \leq y) = \frac{1}{\sqrt{\pi}} \int_{-y}^{y} e^{-\frac{z^2}{2}} dz$$

The Normal Distribution, the Z Distribution, and Intervals

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$

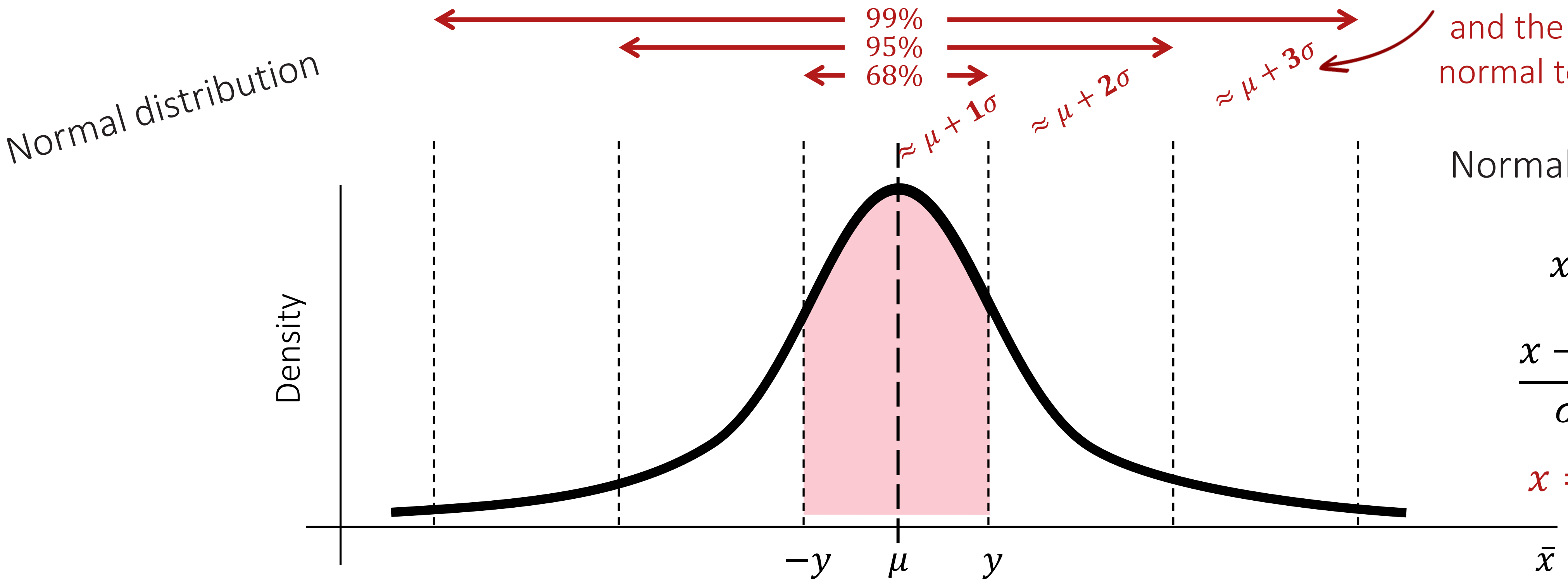


$$P(-y \leq \bar{z} \leq y) = \frac{1}{\sqrt{\pi}} \int_{-y}^{y} e^{-\frac{z^2}{2}} dz = 0.95 \text{ for } y = 1.96$$

This integral has no closed form, but has been tabulated

The Normal Distribution, the Z Distribution, and Intervals

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$



Come from the Z distribution, and the relation linking the normal to the Z distributions

Normal ← Z distribution

$$x \sim N(\mu, \sigma)$$

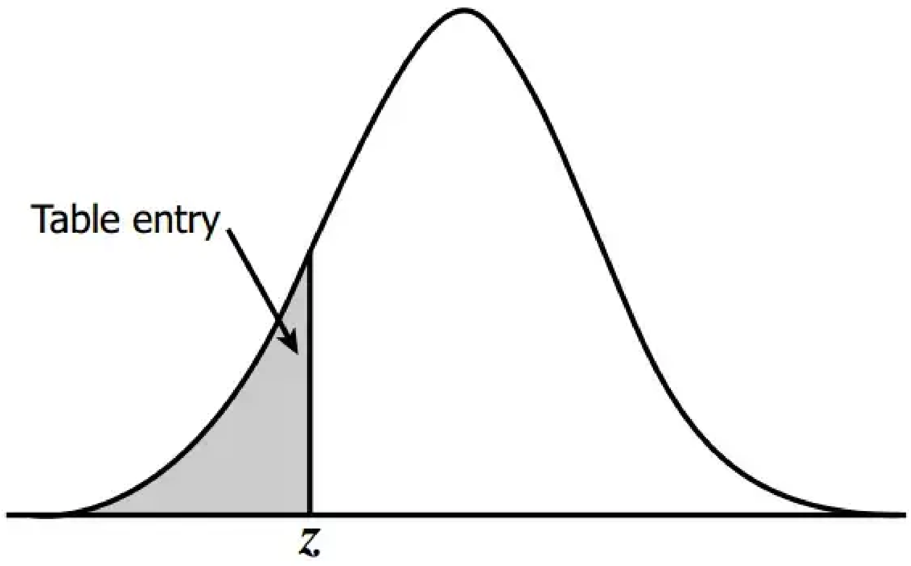
$$\frac{x - \mu}{\sigma} \sim N(0,1)$$

$$x = z \times \sigma + \mu$$

$$P(-y \leq \bar{x} \leq y) = \frac{1}{\sigma\sqrt{\pi}} \int_{-y}^{y} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The Normal Distribution, the Z Distribution, and Intervals

z-table.com

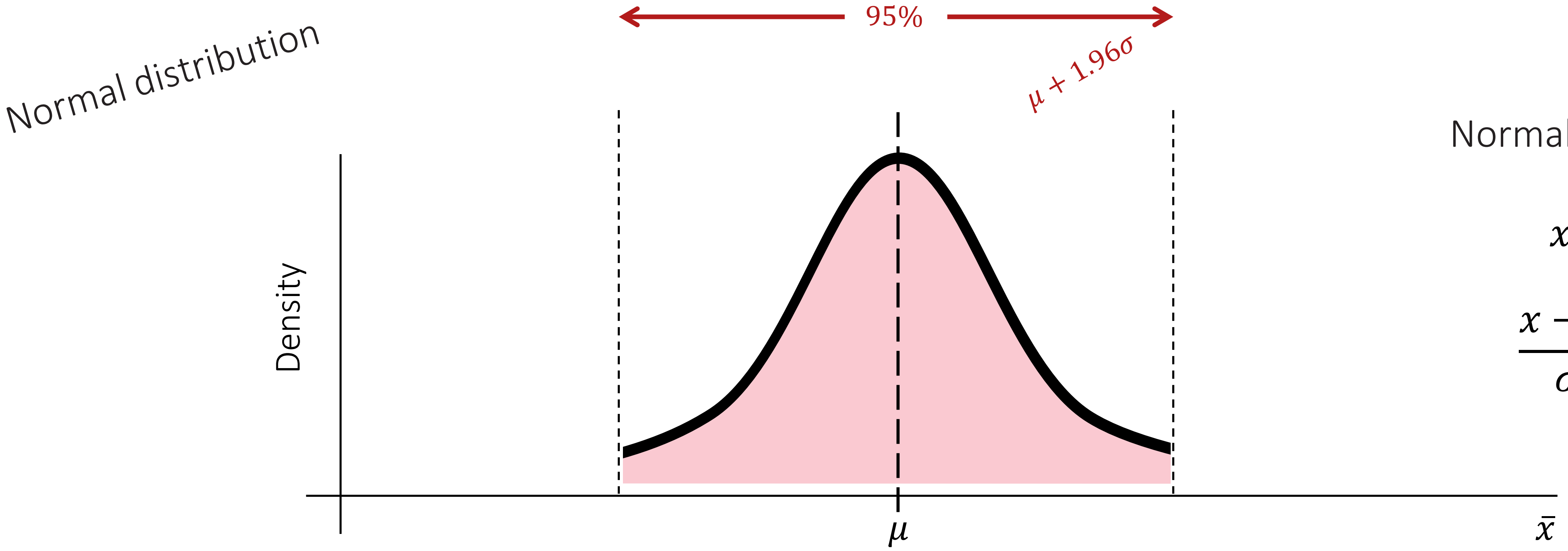


$$P(z \leq y) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{y} e^{-\frac{z^2}{2}} dz$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
–3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
–3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
–3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
–3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
–3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
–2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
–2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
–2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
–2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
–2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
–2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
–2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
–2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
–2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
–2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
–1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
–1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
–1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
–1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
–1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
–1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
–1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
–1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
–1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
–1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
–0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
–0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
–0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
–0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
–0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
–0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
–0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
–0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
–0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
–0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

The Normal Distribution, the Z Distribution, and Intervals

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$



Normal → Z distribution

$$x \sim N(\mu, \sigma)$$

$$\frac{x - \mu}{\sigma} \sim N(0,1)$$

Confidence Intervals: Interpretation

- The same procedure applies to means and proportions

Mean

$\mathrm{CI} = [\bar{\boldsymbol{x}} - \boldsymbol{z}\frac{\boldsymbol{\sigma_s}}{\sqrt{\boldsymbol{n}}}; \bar{\boldsymbol{x}} + \boldsymbol{z}\frac{\boldsymbol{\sigma_s}}{\sqrt{\boldsymbol{n}}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, n the sample size and z the z-score for the confidence level

E.g., age at death in a population

Proportion

$\mathrm{CI} = [\bar{\boldsymbol{p}} - \boldsymbol{z}\sqrt{\frac{\bar{\boldsymbol{p}}(\boldsymbol{1}-\bar{\boldsymbol{p}})}{\boldsymbol{n}}}; \bar{\boldsymbol{x}} + \boldsymbol{z}\sqrt{\frac{\bar{\boldsymbol{p}}(\boldsymbol{1}-\bar{\boldsymbol{p}})}{\boldsymbol{n}}}]$

with $\bar{p}$ the sample proportion, n the sample size and z the z-score for the confidence level

E.g., number of infected people in a population (prevalence)

Side Note: Degrees of Freedom

Consider degree of freedom (df) as exactly what it sounds: the freedom that you have to choose.

Let's take an example, say you are a team coach in a game with 10 players and you have to assign each player to a strategic spot; once you have assigned 9 players to their respective spots, you lose the choice of assigning the 10th, both as a player or as a position. In statistics, this situation is said to have 9 degrees of freedom (10−1).

Another example: Say, I am given a mean of 9 for a set of 3 numbers, At best, I can choose any 2 of the 3 numbers freely but as soon as I do that, I shall lose the freedom to choose the last number—(2+10+x)/3 = 9; x can only be 15.

In other words, df can be defined as the minimum number of independent coordinates that can specify the position of a system correctly.

quora.com/Statistics-academic-discipline-What-is-an-intuitive-explanation-of-degrees-of-freedom

Take Home Messages

B-SIMPLE

- **Biological question and research hypothesis** What am I actually trying to answer?
- **Sample and assumptions** Which approach can I reasonably use?
- **Inference power and significance** Can we, and do we, detect an association?
- **Multiple comparisons and adjustments** Do the thresholds need to be adjusted for multiple testing?
- **Practical effect** What is the magnitude of the association?
- **Logic and check** Does the output make sense? Are the test assumptions verified?
- **Explanation** How do I communicate the results to my audience?

Check your output!!!

Before Next Time (Friday February 20, in LH3)

- Practice: Homework #04 due by **Thursday 5 PM**
- Use the Guiding Quiz #04 if needed

  Some of you should have received messages
- Check you are ready for next class: Knowledge Check #04 due by **Friday 11 AM**

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
- Come to office hours
  - **Tuesdays, 09:00-10:00 AM** (Jordan, Maddy) – Zoom possible
  - **Wednesdays, 05:00-06:00 PM** (Mainul, Princal, Daniel and Harsh) – Zoom possible

  If you cannot make it at these times, please contact the course assistant of your choice by email

  In S1-122, unless indicated otherwise on Canvas